Measurement of Occupancy and Crowding in Embedding Spaces

Sunil Pedapudi*

July 2026

Abstract

Embedding-based systems — retrieval, retrieval-augmented generation, deduplication, clustering — depend on a single geometric property: items that must be distinguished must remain distinguishable under the similarity metric. We call the failure of this property *crowding*. It is a local loss of resolution, invisible to the global statistics that practitioners routinely examine. This report describes `ambit`, an instrument that measures how a corpus occupies its embedding space. It treats an embedded corpus as a pattern of points on a sphere and asks the question other sciences ask of point patterns: do points bunch more than a stated null allows, at which scales, and which points?

The instrument answers the question in its own terms. It measures occupancy at every scale at once, because binned views change with the bin. It compares every measurement to a stated reference — the same corpus with no crowding — because a raw similarity number means nothing on its own. And it carries the diagnosis down to individual documents, because the useful output is a list of what to fix.

The measurements are stated in retrieval terms. A query never lands exactly on its target, and when two documents sit close together, a query aimed at one can return the other — a *collision*. From pair distances alone, `ambit` computes how many collisions to expect at any level of query imprecision, and reports the largest imprecision the corpus absorbs before collisions begin: the *resolution bandwidth* σ^*, the corpus's tolerance for imprecise queries.

Validation establishes three properties. On hundreds of corpora with nothing planted, the instrument alarms only at its advertised rate. It recovers planted groups of near-duplicates in full — groups that mean cosine, isotropy, and hubness miss. And its predictions were tested blind: across four public retrieval collections and seven public encoders, every score was computed and locked before the relevance labels — the human judgments of which documents answer which queries — were consulted. The per-document score picked out in advance the documents that retrieval then failed on, in every pairing that tests it. The corpus-level score tracks encoder quality only on similarity-matching tasks.

The case study embeds one million documents drawn from a public legal corpus; there, the measurements select interventions and verify them. Merging exactly the near-duplicate documents the readouts name, at the similarity level they certify as true duplication, improved twelve queries and harmed none ($p = 0.0005$); merging at the looser level the readouts warn against degraded retrieval. Measurement-built guardrails halve the damage supervised fine-tuning does outside its training distribution, at no cost to its gains. And a rejection rule fixed in advance caught a defective fine-tuning round that its own falling loss endorsed; the labels later confirmed the veto. `ambit` is unsupervised — no labels enter any measurement — and its core depends only on `numpy`.

*The ambit project. Code and reports: `https://github.com/pedapudi/ambit`.

1 Introduction

An embedding model maps items to vectors so that semantic relatedness becomes geometric proximity. Once a corpus is embedded — typically onto the unit sphere $\mathbb{S}^{d-1}$, with similarity the cosine of the angle between items — every downstream capability reduces to one primitive: *given a direction, list the items near it.* The primitive degrades in a specific way when distinct items land too close together: queries cannot separate them, and the system exhibits impostors at the top of rankings, near-duplicates that cannot be ordered, and clusters that blur into one another. We refer to this failure mode as *crowding* and to its complement — the corpus's ability to keep distinct items distinct — as *resolution*.

An embedded corpus is a *pattern of points on a sphere*, and crowding is a question about the arrangement of those points: do some of them bunch together more tightly than a well-spread corpus would allow, at which scales, and which points are they? Questions of exactly this form are the daily business of sciences that live on point patterns — ecologists ask them of trees, epidemiologists of disease cases, astronomers of galaxies — and those fields spent half a century building statistics that answer them at every scale at once, against stated null models, without bins. ambit's design position is that embedding geometry is the same measurement problem on a new domain, and the report carries that methodology to the sphere and gives its outputs retrieval meaning (§3 records the transfers).

Seen this way, crowding is difficult to measure for two reasons, which §2 and §5 develop: it is *local* (a collapsing pocket barely moves any global average), and the standard instruments for localizing density — histograms, hexbin lattices, voxel grids, fixed-k neighbor counts — carry hidden discretization parameters whose values change the answer. Figure 1 previews both difficulties.

This report describes ambit, an instrument built to measure occupancy and crowding without relying on global averages or on a discretization. The instrument answers the point-pattern question in its own terms. It measures occupancy at every scale at once, because any binned view lets the bin choose the answer (§5). It compares every measurement to a stated reference — the same corpus with no crowding — so that the benign concentration every learned encoder exhibits is not mistaken for pathology (§2). And it carries the diagnosis down to named documents, because the actionable output is a list of what to fix, not a score. §2–§8 build and validate the instrument under these commitments; §10 and §11 use its measurements to drive training, deduplication, and fine-tuning guardrails. The report's contributions are the following.

1. **A continuous, null-calibrated measurement layer** (§6), four readouts deep:
    - the exact pair-closeness curve — for each similarity level t, the fraction of document pairs whose similarity is t or higher, a full curve rather than a single count — read against two reference models of a well-spread corpus;
    - a single number summarizing deviation from that reference at all scales simultaneously;
    - a per-item crowding radius with a stability guarantee;
    - a threshold-free catalogue of tight document groups.

    Every component is exercised on one benchmark with a known planted defect (a 5% near-duplicate pocket), so the layers can be read against one another.

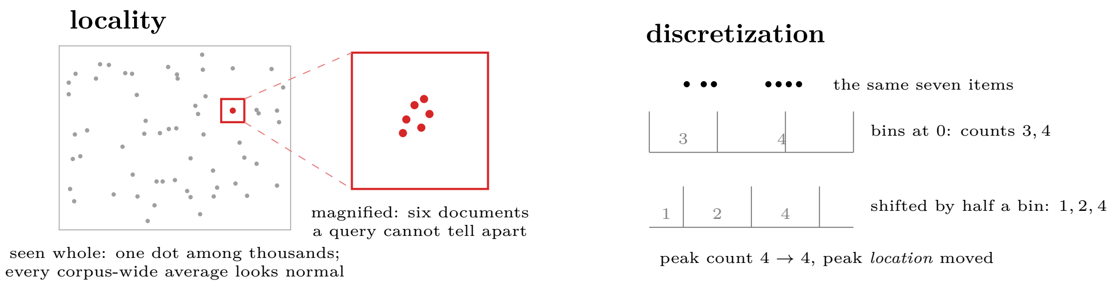

Figure 1: The two measurement difficulties, previewed (schematic). Left: viewed whole, the corpus shows nothing unusual — the defect is one dot among thousands, and on the benchmark of §4, planting 200 near-duplicates in 4,000 items moves mean pairwise similarity by only ≈ 0.002 on a scale of 1 (§2.3). Magnified, that dot is six documents a query cannot tell apart, each expecting 2.6 retrieval collisions against zero for a well-spread corpus. The pathology is severe, local, and invisible to any average. Right: binning the same seven items with the same bin width but a half-bin origin shift changes the counts and relocates the peak — the discretization, not the data, chose the answer. §5 quantifies this on the tool's own displays.

2. **Operational semantics** (§7). An exact formula converts the distance between two documents into the probability that an imprecise query aimed at one returns the other — a collision. Summed over the corpus, this yields the resolution bandwidth σ^*: the corpus's tolerance for imprecise queries. We also show that a widely used training objective, the uniformity loss of Wang and Isola [7], bounds the same collision count at one specific temperature, which gives that parameter concrete meaning. The section closes with a precise statement of the model's assumptions and two negative results recorded with the reasons for their failure.

3. **An open instrument and a training recipe** (§9, §10): a streaming numpy-only measurement core, a case study of one million documents drawn from a public legal corpus, and training penalties built from the measurements, with a guarantee that their gradients act only on confusable pairs and provably leave distant pairs untouched.

4. **Measurement-driven interventions**: (i) a fine-tuning loop (§10.3) in which a rejection rule, fixed in advance and applied to held-out measurements, vetoed a defective round whose own loss curve looked healthy — a veto the labels later confirmed; (ii) label-free deduplication (§10.5) that merges exactly the near-duplicate documents the readouts name, at the similarity level they certify as true duplication — twelve queries improved, none harmed ($p = 0.0005$) — while the same merge at the looser level the readouts warn against degrades retrieval; and (iii) guardrails (§10.6) that halve the damage supervised fine-tuning does outside its training distribution, at no cost to its gains, with a label-free rule for choosing between trained models.

5. **External validation with a measured scope law** (§11). We embedded four public retrieval collections with seven public encoders — twenty-eight combinations — and computed and locked every readout *before* consulting the collections' relevance judgments, the human answer keys that say which documents answer which queries.

The per-document collision score picked out in advance the documents that the judgments then showed failing at their own queries, in every pairing that tests it (sign test $p \approx 2 \times 10^{-7}$). The corpus-level summary σ^* predicts retrieval quality across encoders only where the task itself is similarity matching; on knowledge-seeking tasks the relationship inverts. Together these results fix the instrument's scope: per-document triage and comparative budgeting, not encoder leaderboards.

Throughout, ambit is *unsupervised by design*: no labels, gold pairs, or relevance judgments are consumed. We are explicit about the nature of the contribution: the mathematics assembled here is, with small exceptions, classical, drawn from spatial statistics, directional statistics, geometric inference, and discrepancy theory (§3). The contribution is an *instrument* — the selection, calibration, and composition of known functionals into a measurement system with stated guarantees, stated nulls, and stated scope — together with the negative results that pruned its design space.

2 Geometry of Crowding

This section develops the geometric considerations that the remainder of the report formalizes. Three elementary observations, taken together, determine the measurement design.

2.1 Items are directions; retrieval neighborhoods are caps

Cosine-compared embeddings live on the unit sphere: each item is a *direction*, and similarity is the angle between directions. A retrieval neighborhood — "everything scoring above 0.8 against this query" — is a *spherical cap*: the disc of directions within a fixed angle of the query (Figure 2). This picture reduces every occupancy question to a statement about caps. Corpus concentration corresponds to caps holding more items than their share of the sphere's area; an item's degree of crowding, to the smallest cap around it that already contains other items; and retrieval confusion, to items sharing a small cap — *future collisions*, since a query landing near one cannot avoid the others.

2.2 In high dimension, "random" means "orthogonal"

Fix one item on $\mathbb{S}^{d-1}$ and picture the set of directions at least 0.3-similar to it: a cap around it (Figure 2). In low dimension that cap is a respectable patch of the sphere. As dimension grows, the sphere's area migrates toward the thin band 90° from *any* fixed direction — there are simply ever more ways to be different — and the cap's share of the surface collapses: at $d = 768$, the cap at $\cos \geq 0.3$ holds roughly 10^{-16} of the sphere's area. A uniformly placed second item essentially never lands in it. Quantitatively, the pair cosine of independent uniform directions concentrates as $\mathcal{N}(0, 1/d)$; at $d = 768$ the standard deviation is $1/\sqrt{768} = 0.0361$, and our measured value on 4,000 sampled uniform points is 0.0361 (Figure 3). Three consequences follow. First, "similar" is a geometrically tiny target: two unrelated items land in each other's $\cos = 0.3$ caps about once in 10^{16} draws. Second, the well-spread null's small-distance tail is *empty*. In each of 19 simulated uniform corpora of this size, the fraction of pairs with $\cos \geq 0.30$ is exactly zero, so *any* observed mass there is unambiguous structure. (Nineteen is the classical envelope count: real data exceeding all

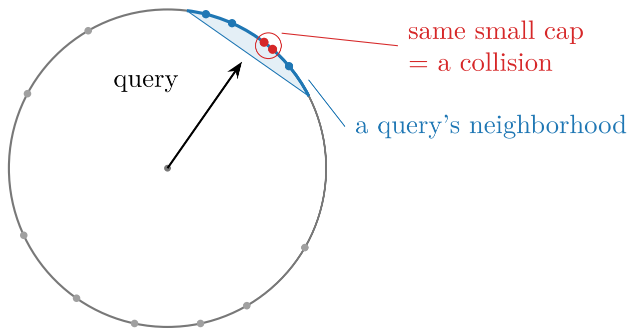

Figure 2: Intuition schematic: items are directions on the unit sphere; a retrieval neighborhood is a cap. Items inside one small cap (red pair) are interchangeable to any query landing there — the geometric definition of a retrieval collision.

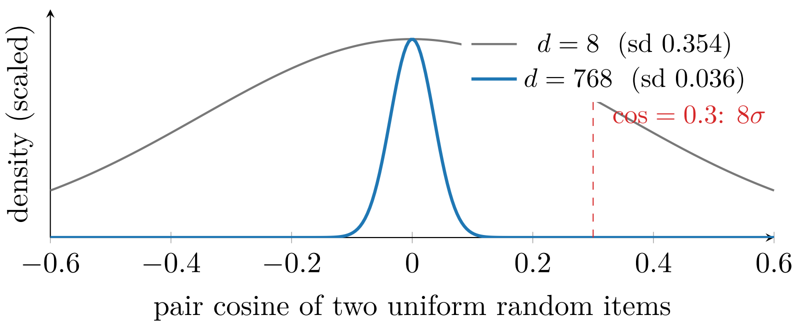

Figure 3: Concentration of measure (analytic densities; the $d = 768$ width is the *measured* sd 0.0361 from 4,000 sampled uniform points, matching $1/\sqrt{d}$ to four decimals): the null pair-cosine distribution collapses to a needle at 0 as dimension grows. The informative region of the null is empty — and the average is dominated by the needle.

19 simulations is a $1/20 = 5\%$ event under the null.) Distance concentration, the classical curse of high dimension [40], is, for a null-calibrated instrument, an asset: every close pair is a finding. Third, because almost every pair is a near-orthogonal background pair, *averages over pairs are dominated by the background* — the observation the next subsection makes precise.

2.3 Why averages cannot see crowding

Every corpus-wide average of a pairwise quantity lives on the *square of all pairs*: an $n \times n$ grid with one cell per ordered pair of items (Figure 4). A group of crowded items occupies a stretch of the square's *side*; its pairs occupy the small square block where that stretch meets itself. Side and area scale differently — membership is linear, pair share is quadratic — so a pocket holding 5% of the items owns only 0.25% of the area over which any average is taken. The benchmark corpus used throughout this report makes the dilution concrete. It holds $n = 4{,}000$ items at $d = 768$, and the last $k = 200$ of them — five percent — are replaced with near-duplicates of one another, so alike (median pairwise cosine 0.929) that a retrieval system cannot rank them apart (full construction in §4). Now follow the arithmetic of the

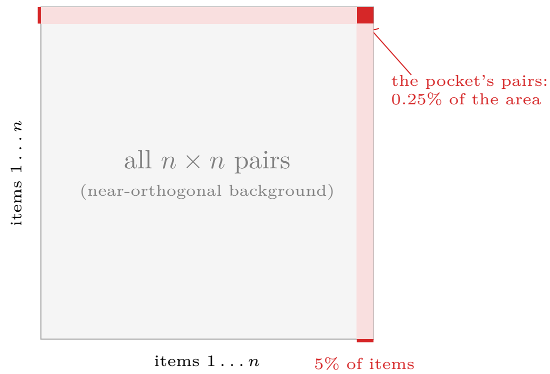

Figure 4: The square of all pairs (schematic; percentages from the d=768 benchmark). Every average of a pairwise quantity is taken over this square's area. Crowded items occupy a stretch of the *side*; their mutual pairs occupy only the small block where that stretch meets itself — 5% of the items, 0.25% of the pairs. The pale strips are the pocket's pairs with the bulk, which are near-orthogonal like the background. Averages weight by area, so a catastrophic pocket is geometrically diluted before any arithmetic begins.

average, step by step. Five percent of the items yields $0.05^2 = 0.25\%$ of the pairs, so 99.75% of the average consists of background pairs, which the previous subsection showed sit near cosine 0. Each pocket pair contributes its cosine of roughly 0.93, weighted by the pocket's 0.25% pair share, so the entire catastrophe adds $0.0025 \times 0.93 \approx 0.002$ to the mean. Two hundred documents became interchangeable, and the corpus-wide average moved from about 0.000 to 0.002: invisible without a calibrated test, and uninformative about *which* items failed. Covariance-spectrum summaries (effective rank, isotropy scores) are blind for the same arithmetic reason: they are sums over the corpus, dominated by the diffuse majority. More generally: *global averages address whether the whole cloud is degenerate; crowding concerns which part of the cloud has collapsed, at what scale, and which items it contains.* A sound instrument must therefore (i) read *distributions*, not means; (ii) read them *at every scale*, since pockets live at small scales the bulk never visits; (iii) *calibrate* against a matched null, since absolute cosine is not self-interpreting; and (iv) descend to *named items*, since the actionable output is a list of ids. These four desiderata are the design of §6.

3 Related Work

The introduction framed an embedded corpus as a point pattern and crowding as the clustering question that the observational sciences ask of point patterns. Those sciences did not stop at the question: they spent decades building the statistics that answer it, and ambit is rooted in that body of work, carried to the high-dimensional sphere where embedded corpora live. This section describes each borrowed method in plain terms — what it was built to do in its home field, and what changes when it is applied to embeddings — and then situates ambit among the statistics the embedding literature itself uses. Spatial statistics supplies the crowding curve and its inference; geometric inference, the per-item crowding radius and the pocket hierarchy; discrepancy theory, the occupancy summary; directional statistics, the reference that models the cone. What no donor field supplies is the operational layer: none has a notion of one document outranking another at a query, so the conversion of pair distances into expected retrieval collisions (§7) is native to this report.

Spatial statistics. Ripley's K function [16, 17] is the ecologist's tool for asking whether points (trees, stars, cells) cluster: count the pairs within every distance and read clustering as a *function of scale* rather than at one arbitrary radius. ambit's crowding curve is this idea transplanted to the sphere, where the null is a cap-area law and, the domain having no boundary, the planar edge corrections vanish [18]. Two further pieces of the field's machinery carry over with the curve. Global envelope tests [21] answer the companion inference question — how to test a whole curve at once without the multiple-comparison error of checking each scale separately — and supply ambit's liftoff test. And spatial statisticians long ago met ambit's central confound: raw K misreads a nonuniform background intensity as clustering, and the *inhomogeneous* K function [19] repairs this by conditioning the test on a fitted intensity. The anisotropy-conditioned reference of §6.1 is the same move on the sphere, with the corpus's cone playing the role of the fitted intensity and the angular central Gaussian of directional statistics supplying the fitted model.

The hazards of partitioned statistics. The same sciences also documented what *not* to transfer. The modifiable areal unit problem (MAUP) [24, 25, 26] and the ecological fallacy [27] are the classical demonstrations that partition-based statistics change with the partition and say nothing about individuals — the two failures, quantified on embedding displays in §5, that ambit is built to avoid. The same lesson appears in one dimension as the histogram's bin-width and origin sensitivity [28, 29, 31]; the averaged shifted histogram [30] repairs placement by averaging over origins, and ambit goes one step further by removing the partition altogether.

Discrepancy theory. Stolarsky's invariance principle [22, 23] converts an audit that cannot be run directly — compare actual against expected occupancy over *every* cap at *every* scale — into one computable number, the mean pairwise distance. ambit's occupancy discrepancy is this theorem put to work.

Geometric inference. The distance to a measure [32, 33] asks, for each point, how far it must reach to gather a fixed fraction of the data — a robust replacement for density estimation. Its defining property is stability: small changes to the data provably move every score only slightly. ambit's per-item crowding radius is this estimator on the sphere. For structure, Hartigan's cluster tree [37, 38] reads groups at all densities simultaneously instead of at one threshold; ToMATo [34] and HDBSCAN's condensed tree [35] are two normalizations of that object [36], with stability inherited from persistence theory [39]. ambit's pocket catalogue — births, deaths, and prominences of tight groups — is this machinery with a null-calibrated reporting floor.

Anisotropy and resolution. The embedding literature contributes the phenomenon that every one of the transfers above must be conditioned on. Gao et al. [1] and Ethayarajh [2] established that learned embeddings do not use the whole sphere: vectors bunch into a narrow cone, which inflates every pairwise cosine at once. For ambit this is the central confound — a cone raises similarities without making any two items confusable — and it is why every readout is calibrated against an anisotropy-matched reference rather than against the uniform sphere alone. Mu et al. [3] showed that subtracting the mean direction and a few dominant components restores much of the usable angular range: this is precisely the linear repair that ambit's routing recommends when its measurements attribute crowding to the cone and to nothing else. Cai et al. [4] found genuine cluster structure *inside* the cone — the kind of beyond-cone crowding the anisotropy-matched reference is designed to isolate. Effective rank [5] and IsoScore [6] compress the covariance into one number (roughly, how many directions carry variance, and how close the cloud is to perfectly round); ambit reports both as context and measures, in §8, that they cannot see small pockets. Wang and Isola [7] decompose contrastive embedding quality into *alignment* (paired texts land together) and *uniformity* (everything spreads out): ambit deliberately measures only the uniformity half — it is unsupervised and cannot see alignment — and this decomposition is exactly what the external validation of §11 recovers empirically. Hubness [8] is the phenomenon of a few items appearing in disproportionately many other items' nearest-neighbor lists — a metric-level symptom of the same congestion ambit measures geometrically, and a baseline in the detection-power study.

Embedding-quality scores. RankMe [9], α-ReQ [10], and their relatives (surveyed in [11]) score an embedding, without labels, by how evenly its variance spreads across directions. A structural fact governs the family: on unit vectors these scores all re-read the same second-moment spectrum, so they largely re-parameterize one another. The Vendi score [12, 13] escapes the family by counting effective modes through a nonlinear kernel [14, 15]. ambit adopts the lesson as a design filter (§6.5): rather than adding another reading of the spectrum, it measures different objects entirely — the pair-distance distribution and per-item fields.

Projections and density. Two-dimensional layouts preserve neighborhoods, not density [43, 44]: a visually dense region of a projection need not be dense in the embedding, which is why ambit draws projections but never measures on them. Kernel density estimation converges at $O(n^{-4/(4+d)})$ — vacuous at d in the hundreds [31] — which is why the per-item layer rests on the distance to a measure instead. Local intrinsic dimension estimators [41, 42] are used only per-item and comparatively.

4 Experimental Methodology

Every claim marked *measured* in this report follows one protocol, stated here once. The test environments are scientifically incidental and are described by their properties in Appendix C; the data are what matter.

Datasets. Three complementary data regimes carry the evidence; among them, every claim is testable.

(i) The synthetic benchmark family — known ground truth by construction. Detection, localization, and recovery claims can only be scored against a pathology whose membership and scales are known exactly; planting one supplies that ground truth as *design knowledge*

Table 1: Evaluation corpora. All are public; the first four are used in their BEIR form [51].

corpus	documents	judged queries	task
TREC-COVID [52]	171,332	50 (dense, graded)	biomedical search
Quora [48]	522,931	10,000	duplicate matching
Natural Questions [53]	2,681,468	3,452	open-domain QA
ESCI / Shopping Queries [54]	1,215,854	8,956 (graded)	product search
Nemotron-Legal [60]	1,000,000	1,000 + 1,496	statute/case retrieval

without the instrument ever consuming labels. The traced benchmark of §2.3 is: $n = 4{,}000$ points at $d = 768$ (a contemporary encoder width), drawn as normalized standard Gaussians (exactly uniform on $\mathbb{S}^{d-1}$); the pocket replaces the last $k = 200$ points (5%) with $x_i = (c + s\,g_i)/\|c + s\,g_i\|$ for one shared uniform direction c, i.i.d. Gaussian g_i, and spread $s = 0.01$. This spread places the pocket in the near-duplicate regime (median intra-pocket cosine 0.929: items a retrieval system cannot rank apart) while leaving every global average essentially unmoved (§2.3), the property that makes it a stringent test for a local diagnostic. The uniform control is the same draw without the replacement. The wider family varies what the traced instance fixes: pocket count (1–20), size (20–800), and tightness ($s = 0.005$–0.06, spanning near-duplicate to loose association); mean-shift cones; low-rank collapses; and pure nulls — the grid behind the sensitivity, calibration, and power studies. All generators are stated in full above or beside their experiments; nothing is tuned to the instrument.

(ii) A public real corpus — realistic geometry at scale. Synthetic corpora cannot exhibit the learned-embedding pathologies that motivate the instrument, so the case study (§9) uses *Nemotron-Pretraining-Legal-v1* [60], a public (CC-BY-4.0) legal corpus published by NVIDIA: 10^6 documents drawn from four of its thirteen subsets — the California Code of Regulations, case-law summaries, CaseHOLD, and eCFR question–answer pairs — embedded at $d = 1024$ by the Qwen3-Embedding-0.6B instruction-tuned encoder [47], L2-normalized, stored as sharded Parquet with stable document ids. It supplies what the benchmark cannot: a genuine anisotropy cone (mean pair cosine +0.236), *diffuse* within-subset crowding rather than planted pockets, subset labels used only to *condition* one readout (never consumed by a measurement), and corpus scale that exercises the streaming path.

(iii) A public evaluation grid — external outcomes. The external study (§11) and the retrieval-facing experiments (§10.5–§10.6) additionally use four public retrieval collections with relevance judgments and seven public encoders, enumerated with citations in Tables 1 and 2. Judgments are quarantined: every readout is computed and hash-frozen before any judgment file is opened, and judgments are consumed exclusively by scoring code. The three regimes are deliberately complementary: the benchmark makes claims falsifiable, the corpus makes them representative, and the grid makes them externally scored.

Standard settings. The following parameters govern statistics defined in §6–§7, each introduced fully where its statistic is defined. Unless an experiment sweeps them: pair statistics use 2×10^5 sampled distinct pairs; whole-curve inference uses the global rank-envelope test at 199 null curves (99 inside report figures); the anisotropy-conditioned reference is sampled from the corpus's own covariance eigenvalues at matched pair count; the DTM (distance-to-measure) per-item crowding field uses mass fraction $m = 0.02$; the merge tree uses minimum

Table 2: Encoders. Dimensions are output embedding widths; all corpora are embedded with each model's documented document/query prompt conventions.

encoder	params	dim	lineage
all-MiniLM-L6-v2 [55, 49]	22M	384	distilled BERT
all-mpnet-base-v2 [55, 50]	110M	768	MPNet
E5-base-v2 [56]	110M	768	weak-supervision contrastive
Arctic-Embed-m-v1.5 [58]	110M	768	retrieval-specialized
EmbeddingGemma-300m [59]	300M	768	Gemma
BGE-large-en-v1.5 [57]	335M	1024	C-Pack
Qwen3-Embedding-0.6B [47]	0.6B	1024	instruction-tuned decoder

pocket size 8; the discrepancy z uses 64 null replicates. Each of these defaults is itself validated by a sweep reported in the section that uses it.

Per-experiment protocols. Each experiment below is stated as: the hypothesis under test; the design, with the reason it takes that shape; the control it is read against; and the measure that decides. Full parameter values appear beside each result.

Lattice fragility (Table 3).

- **Hypothesis:** the headline statistics of binned displays are properties of the lattice, not of the data.
- **Design:** hold 4,000 synthetic 2-D points fixed and perturb only the lattice — shift its origin by fractions of one cell, change its column count — because any resulting change can then be charged to the partition alone.
- **Control:** an isotropic jitter of the *data* by 0.1% of their span, the honest magnitude of change a data perturbation of comparable size produces.
- **Measure:** the change in the display's peak cell count and peak location under each perturbation, lattice against control.

Confusion-law verification (Table 5).

- **Hypothesis:** the closed-form probability that noise flips a ranking is exact in any dimension.
- **Design:** two fixed unit vectors at a known angle, 3×10^5 independent noise draws per cell of a 6×4 grid of angles and noise scales — direct simulation of the law's own setting, with no modeling assumptions to hide in.
- **Control:** the closed form itself; under the hypothesis the residual is pure Monte-Carlo noise of predictable size.
- **Measure:** the maximum absolute deviation between empirical flip rate and formula over the grid.

Redundancy of the confusion and uniformity functionals (Table 6).

- **Hypothesis:** the two functionals may order corpora identically, making one redundant.
- **Design:** eight synthetic corpus generators spanning the pathology families, both functionals computed on each — redundancy is a claim about *orderings across corpora*, so the design produces orderings, not values.
- **Control:** perfect rank agreement, the outcome under full redundancy.
- **Measure:** the rank correlation and, more importantly, each disagreement, inspected individually.

Calibration and detection power (§8).

- **Hypothesis:** the instrument's alarms fire at their nominal rates on healthy corpora and detect planted pathologies that standard summaries miss.
- **Design:** 400 pure-null corpora first, to measure false-alarm rates and to set every detector's threshold at the same 1% specificity — power comparisons are meaningless unless all detectors are equally trigger-shy — then a grid of planted pathologies, 50 replicates per cell, with threshold seeds disjoint from test seeds.
- **Control:** the pure nulls, and the standard summaries (mean cosine, isotropy scores, effective rank, hubness, a neighbor-distance quantile) run under identical thresholds.
- **Measure:** empirical false-alarm rate against nominal; detection rate per pathology cell at matched specificity.

Sensitivity sweeps.

- **Hypothesis:** every default parameter sits on a plateau, so reasonable variation does not change conclusions.
- **Design:** sweep each knob (reservoir size, pair-sample size, mass fraction, minimum pocket size, null-replicate count) one at a time over an order of magnitude or more.
- **Control:** the readout at the default, with seed-to-seed spread as the yardstick.
- **Measure:** readout drift across the sweep, compared to that spread.

The training example (Table 10).

- **Hypothesis:** penalties built from the measurements reduce retrieval collisions without destroying the embedding's structure.
- **Design:** a planted pocket with a linear map as the smallest possible trainee, and a 20% slice held out from mining, weighting, and batching — verdicts must come from data the training machinery never touched.
- **Control:** the corpus before training, and a structure-preservation score against the frozen base.
- **Measure:** held-out flagged-item collisions and neighbor overlap, per schedule.

The encoder loop (§10.3).

- **Hypothesis:** held-out geometric readouts alone can adjudicate fine-tuning rounds — accepting safe ones and rejecting harmful ones — with no labels.
- **Design:** a fixed stratified 200,000-document subset with 20% held out, and a continuation rule registered before any round ran, so acceptance cannot be negotiated after seeing results.
- **Control:** the base model's readouts; and the trainee's own loss curve, the signal a practitioner would otherwise trust.
- **Measure:** per-round held-out σ^*, flagged-cohort collisions, and neighbor overlap against the rule.

Routing (§10.5).

- **Hypothesis:** a repair succeeds only at the site and scale the measurements prescribe — not because deduplication is generically good.
- **Design:** the same label-blind deduplication executed in all four combinations of two sites (foreign distractors; the target corpus) and two scales (the confusable window; the duplicate scale), because only the full 2×2 separates "the prescription was right" from "any cleanup helps."
- **Control:** the three cells the measurements warn against.
- **Measure:** frozen-query failures fixed and introduced per cell, with an exact paired test.

Guardrails (§10.6).

- **Hypothesis:** measurement-derived guard terms prevent the collateral damage of supervised fine-tuning without reducing its gains.
- **Design:** one supervised recipe run twice, the arms differing *only* in the guard terms, so any outcome difference is attributable to them; the held-out measurement gate is read and recorded before any judgment.
- **Control:** the unguarded arm.
- **Measure:** the targeted retrieval metric (where no cost is claimed) and the untargeted open-corpus metrics (where protection is claimed), plus the pre-label gate's prediction.

The external grid (§11).

- **Hypotheses:** first, that each document's expected-collision count predicts which judged-relevant documents fail at their own queries; second, that the corpus-level σ^* ranks encoders within a corpus.

- **Design:** every corpus–encoder cell is embedded and fully measured, and the readouts hash-frozen, *before* any judgment file is opened — the freeze makes leakage from labels into measurements impossible rather than merely disavowed.
- **Control:** for the per-document claim, the natural geometric baseline (nearest-neighbor cosine); for the corpus-level claim, mean pair cosine.
- **Measure:** per-cell prediction quality (AUC) for the first; per-corpus rank correlation with nDCG@10 for the second.

Reporting conventions. Aggregates are reported with their null value at matched sample size wherever a null exists; effect sizes are stated in the null's units (z-scores) or as ratios. Negative results are reported with the same prominence as positive ones. Headline benchmark numbers are replicated over 20 seeds and reported as mean ± sd; calibration and power studies use the replicate counts stated with them (400 nulls, 50 per pathology cell, 199-null rank envelopes, 64-replicate discrepancy nulls); single-seed values appear only for deterministic or concentration-driven demonstrations and are marked by the absence of a spread. *Figures* follow one rule: every data-bearing plot embeds unmodified experimental outputs as its coordinates.

- **Real-corpus figures:** the crowding curve, the DTM field, and the collision curves plot Nemotron-Legal measurements, and two figures reproduce the instrument's own report readouts on that corpus natively (the named-document localization list and the pockets bar, with the report's values verbatim).
- **Self-computed demonstrations:** the lattice demonstration's highlighted peak cells are computed from the figure's own points.
- **Intuition schematics:** the remaining figures carry no data and are captioned as schematics.

5 The Fragility of Binned Occupancy

ambit's early density views were conventional: hexbin lattices over a 2-D projection, voxel grids, radial shells, fixed-k neighbor counts. This section records, quantitatively, why the measurement layer abandoned them.

5.1 The answer belongs to the lattice

Figure 5 and Table 3 show both MAUP effects [26] on the tool's own display. The *zonation* effect is lattice placement: shifting the lattice by half a cell moves the headline peak count from 310 to 403 and relocates which cell is the peak. The *scale* effect is cell size: across reasonable column counts the peak count varies by a factor of 2.4. Neither perturbation touches the data. The classical demonstration that re-zoning fixed data can move a correlation across essentially $[-1, +1]$ [25] is the extreme form of the same phenomenon. Histograms also lose statistical efficiency to discretization: their mean integrated squared error shrinks as $O(n^{-2/3})$ with sample size, against $O(n^{-4/5})$ for kernel estimators [28, 29]. Averaging

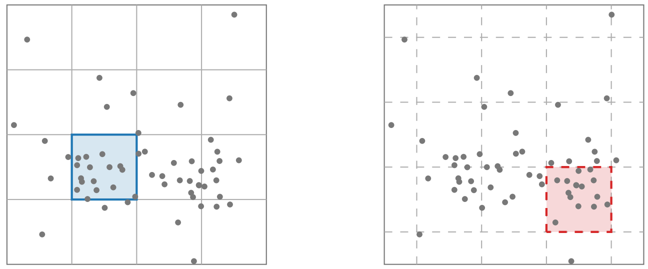

Figure 5: The modifiable areal unit problem on fixed data (the diagram's points and highlighted cells are computed, not sketched): shifting the same lattice by half a cell changes the peak count *and relocates which cell is the peak.* Table 3 gives the effect at scale on the tool's own hexbin display.

Table 3: Lattice fragility on fixed data (n=4,000 synthetic 2-D points; the tool's own hexbin assignment). Perturbing only the *instrument* moves the headline statistics by 30% (placement) and 2.4× (scale); a 0.1%-of-span data jitter, by contrast, moves the peak by one count — fragility is to the lattice, not the data, until a point crosses a boundary.

perturbation	peak-cell count	occupied cells
lattice as placed	310	256
lattice shifted 0.25 cell	310	253
lattice shifted 0.5 cell	403	262
20 columns	574	168
26 columns	340	256
34 columns	242	350
data jittered by 0.1% of span	339	—

histograms over many lattice placements — the averaged shifted histogram [30] — removes the placement dependence and recovers the kernel rate. `ambit` retains one hexbin *display*, averaged over placements in exactly this way and captioned as a view, not a measurement.

5.2 One hidden scale each; duplicate blindness

Crowding is multi-scale: near-duplicate pockets live at tiny angles, topical clusters at moderate ones, the anisotropy cone at large ones. Every bin width, shell count, and fixed neighbor count k silently commits to one scale. Injecting 200 near-identical points — the benchmark's standard planted-pocket size, 5% of its $n = 4{,}000$ — inside an already-dense cell moves the hexbin peak from 340 to 540, indistinguishable from a benign hot cell — while the continuous per-item field of §6.3 collapses by a factor of 18 at its first percentile. A fixed k in a kNN graph is the same defect in rank-based form: in a dense pocket k=10 spans a tiny radius, while in a sparse outer region it spans a huge one. "Neighborhood" therefore denotes a different physical scale at every point — precisely the drift a crowding measure must exclude.

5.3 Steps, nulls, names, and projections

Four further costs are structural.

- *Instability.* A bin count is a step function of the data, with no guarantee on how far a small perturbation can move it. The field of §6.3 is provably Lipschitz in the Wasserstein distance of the input and measured stable (self-correlation 0.997 under jitter at d=768).
- *No null.* A peak count carries no statement of what a well-spread corpus would show; a perfectly uniform cloud still has a hottest cell.
- *No names* (the ecological fallacy [27]). A cell count cannot say which items are crowded, and the actionable output of a diagnosis is a list of ids.
- *Projection.* Binning a 2-D layout measures the projection, not the object: far-apart groups collide in a cell, orthogonal crowding vanishes, and nonlinear layouts actively destroy density [43].

The repair is not a better ambient density estimate — kernel density estimation is vacuous at d in the hundreds [31] — but the abandonment of partitions for scale-free functionals of interpoint distances, whose guarantees carry no dimension factor.

6 Continuous Occupancy

This section presents the measurement layer: four instruments — a whole-curve statistic, a one-number discrepancy, a per-item field, and a structural tree — answering, in order, the desiderata of §2.3. Let $x_1, \ldots, x_n \in \mathbb{S}^{d-1}$ (unit-normalized; cosine $c_{ij} = \langle x_i, x_j \rangle$; chord $r_{ij} = \|x_i - x_j\| = \sqrt{2 - 2c_{ij}}$). ambit computes exact full-corpus second moments by streaming and draws a bounded uniform *reservoir* (a fixed-size random subsample kept during the single pass) plus a large random-pair cosine sample for everything else. The pair-sample default of 2×10^5 is sized so that the scale readouts of this section are stable to a few thousandths; the measured concentration rates appear with the discrepancy statistic below. Throughout this section the *figures* show the real corpus (Nemotron-Legal, §4) — the primary examples are drawn from learned-embedding geometry rather than synthetic constructions — while the traced benchmark of §2.3, whose planted pathology is known exactly, supplies the falsifiable counterpart for every layer in the text.

6.1 The crowding curve

Definition 1 (Pair-closeness curve)**.** $K(t) = \Pr_{(i,j)}[\, c_{ij} \geq t \,]$, *hereafter the crowding curve: the fraction of distinct pairs at least t-similar, estimated* exactly *from the pair sample by sorting — no bins, no origin, no width.*

Recall that Ripley's K function is the spatial statistician's device for asking whether points cluster: for every radius at once, count the fraction of pairs that lie within that radius of each other, and compare the resulting curve against a random scatter. $K(t)$ is this function on the sphere [16], in cosine coordinates. Two references with matched sample size accompany it (Figure 6):

(i) The uniform null, tested globally. For i.i.d. uniform points the pair cosine satisfies $c = 2B - 1$, $B \sim \mathrm{Beta}\big(\frac{d-1}{2}, \frac{d-1}{2}\big)$, so null pair samples are drawn directly — no d-dimensional vectors are materialized. The sphere is boundaryless, so no edge correction is required and none can be misapplied [18]. Inference over the whole curve uses a one-sided global *rank-envelope* test [21]. The test works by simulation: the data curve is ranked among 199 curves drawn from the null, so the smallest attainable p is 1/200, and a single global p covers all scales at once. Ties are broken by extreme rank length, because a one-sided test on K is heavily tied wherever every null curve is exactly zero. *Liftoff* — the highest cosine at which the data exceed the α-critical envelope, the null-curve band at the test's significance level — is annotated only when the global test rejects. On 40 pure-null corpora the gated test reported 0 liftoffs. The naive pointwise band (the max of the null curves) fired on *every* null seed at the bulk edge — a direct demonstration of the multiple-comparison failure — and even the α-critical envelope alone over-fires $\approx 4\times$ nominal. The division of labor is therefore strict: the envelope locates the scale, and the rank p decides significance.

(ii) The anisotropy-conditioned null. Real corpora are cone-shaped for benign reasons; judging them against uniformity alone over-alarms. The second reference is the *angular central Gaussian* (ACG) of directional statistics [20] with the corpus's own covariance spectrum — $x = \Sigma^{1/2} g / \|\Sigma^{1/2} g\|$, sampled from the eigenvalues alone by rotation invariance: the corpus's second-moment cone *without* its clustering. Height of the data curve above this reference is crowding the cone cannot explain.

Figure 6 shows the curve on the real corpus (Nemotron-Legal, §4): the global test rejects at the minimum attainable $p = 1/200$ with liftoff at cos ≈ 0.81. The reading is more detailed than a single scalar: through mid-similarities the data track the ACG reference — that part of the excess over uniformity *is* the anisotropy cone, benign and expected of a learned encoder — while above cos ≈ 0.8 the data rise above both references: genuine tight-pair structure at the near-duplicate scale, which §6.3 will attribute to named documents. On the traced benchmark the same test rejects at $p = 1/200$ with liftoff cos ≈ 0.93 and a shoulder of excess mass exactly matching the planted pocket's pair share (2.4×10^{-3}) — the counterpart with known ground truth. The lower panel of Figure 6 repeats the reading across five corpora under one encoder: every corpus carries tight-pair mass far above the null, but the anatomy differs — ESCI and TREC-COVID hold the most near-duplicate mass, Quora the least beyond its paraphrase band — structure a single scalar would average away.

6.2 The occupancy discrepancy (Stolarsky z)

A binned count inspects one cell of one lattice. The complete version of that question inspects every spherical cap — every center, every radius — and asks how far each cap's occupancy departs from the uniform expectation. A classical identity collapses this total departure into a single pairwise average.

Proposition 1 (Stolarsky [22]). *For points on $\mathbb{S}^{d-1}$, the spherical-cap L_2 discrepancy — the squared deviation between empirical and expected cap occupancy, integrated over* all *cap centers and* all *cap radii — equals, up to explicit constants, a constant minus the mean pairwise chord $\frac{1}{n^2}\sum_{i,j}\|x_i - x_j\|$.*

The mean pairwise chord is therefore "every lattice at every scale at once": the exact completion of what binned occupancy approximates, with no lattice to choose. ambit reports

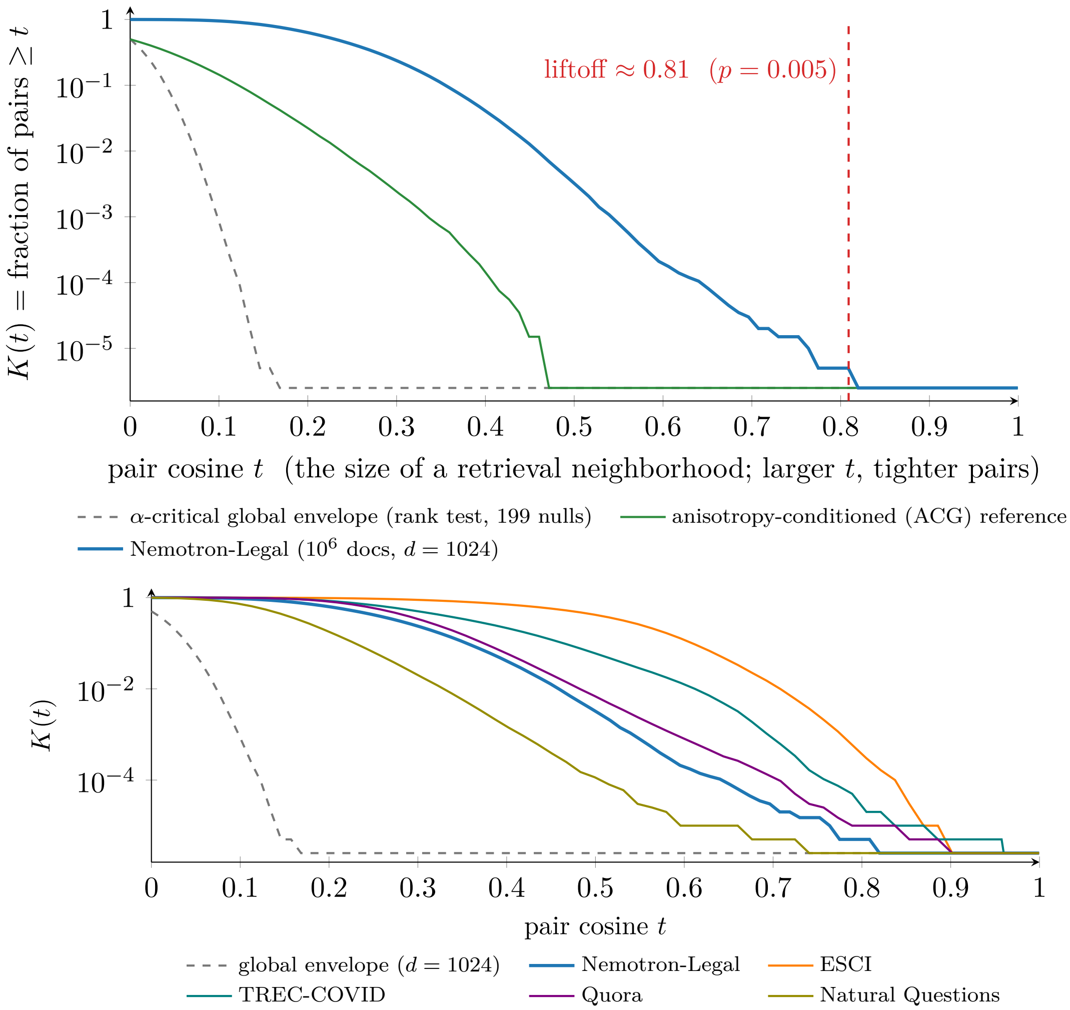

Figure 6: The crowding curve on the real corpus (Nemotron-Legal [60], computed from the report's own $2{\times}10^5$-pair sample): the data track the anisotropy-matched reference through mid-similarities and rise above both references past $\cos \approx 0.8$, where the global test rejects at $p = 1/200$. No bin width or lattice placement exists anywhere in the figure, and the liftoff claim is family-wise valid across all scales. Bottom: the same curve for five corpora under the same encoder ($d = 1024$, shared envelope), each with its own anatomy; the text gives the reading.

it as a z-score against the uniform null at matched pair-sample size, pricing in sampling noise. Measured on the benchmark over 20 seeds: the uniform control scores $z = -0.14 \pm 1.00$ — exactly calibrated — while planting the 5% pocket moves the mean chord from 1.41405 to 1.41160, a shift of 0.17%, which the concentrated null resolves as $z = -47 \pm 4$. Tightening the pocket further (members within a cap of angular radius ≈ 0.15) drives the same statistic to $z = -430$: detection strength scales with the pathology while the control stays at noise level.

Two properties of z are stated so it is read correctly. First, *z is a test statistic, not an*

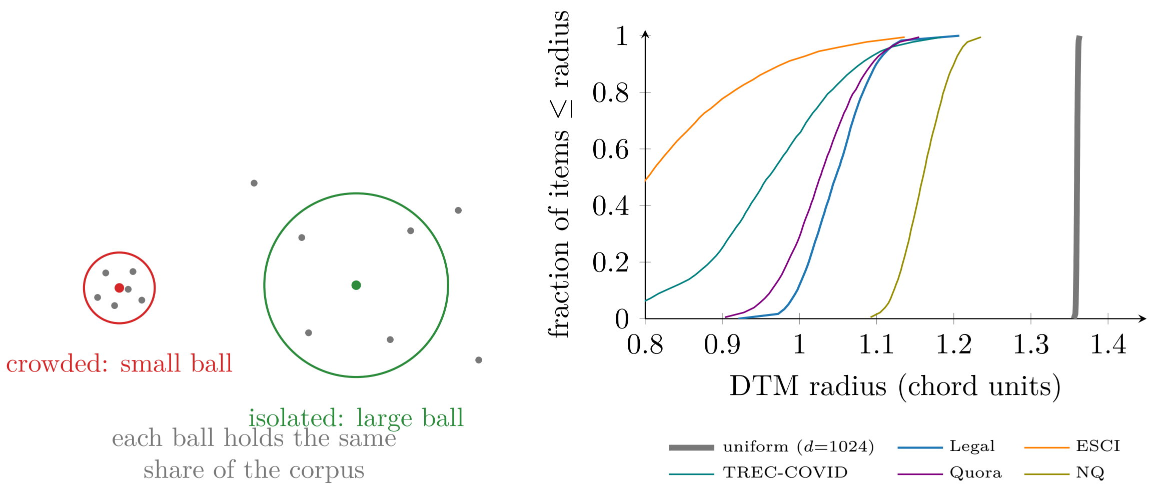

Figure 7: The distance-to-measure field. *Left:* the intuition schematic — each item is scored by the radius of the ball it needs to gather a fixed share ($m = 2\%$) of the corpus. *Right:* the field's exact CDF on the real corpus (measured): the *entire* field sits left of the uniform needle at 1.36, and the finding is the low tail against the corpus's own bulk (1st percentile 0.966 against median 1.047), whose members the instrument names by id (Figure 8); the text explains why the uniform band reads on the cone rather than on pockets. The four grid corpora under the same encoder overlay their CDFs: each has its own bulk position and low-tail mass — per-item crowding profiles differ far more across corpora than any global statistic suggests.

effect size. Its magnitude grows with the pair-sample size P, because the null mean's spread shrinks as $1/\sqrt{P}$. z values are therefore comparable only at matched P, and the instrument reports them at its standard $P = 2{\times}10^5$. The scale readouts σ^* and liftoff behave like effect sizes instead: they concentrate as $1/\sqrt{P}$, with $\mathrm{sd}(\sigma^*)$ falling from 0.0126 to 0.0013 as P grows from 10^4 to 10^6. Second, the null's spread must itself be estimated well: with 24 null replicates $|z|$ is inflated $\approx 20\%$ and reproduces with sd ≈ 8. The default of 64 replicates sits where that inflation has died away; at the default the benchmark reads $z = -43 \pm 4.5$.

6.3 Localization: the distance-to-measure (DTM) field

Definition 2 (Empirical DTM [32, 33])**.** *For mass fraction m and $k = \lceil mn \rceil$:* $\mathrm{DTM}_m(x_i) = \left(\frac{1}{k}\sum_{j=1}^{k} r_{i,(j)}^2\right)^{1/2}$*, with $r_{i,(j)}$ the distance to x_i's jth nearest reservoir point — the radius the item needs to gather an m-share of the corpus.*

A small radius means the item is crowded; a large radius means it is isolated (Figure 7). The single parameter m is a *mass fraction*, scale-free in n, and replaces bin width, grid origin, and fixed k simultaneously. Its one trade-off is quantified in Table 4: separation between a planted pocket and the bulk collapses almost exactly at the predicted blur boundary $m \approx k/n$, so the reporting rule is "choose m below the smallest pocket share you care about," and the default $m = 0.02$ resolves pockets down to 2% of the corpus.

The estimator inherits the stability theorem $\|\mathrm{DTM}_m^P - \mathrm{DTM}_m^Q\|_\infty \le m^{-1/2}\, W_2(P,Q)$ [32]:

Table 4: The DTM mass fraction's measured blur boundary (bulk median / pocket median of the field; benchmark family, planted pocket of k members in $n = 4{,}000$). Separation survives while $m \lesssim k/n$ (boldface) and collapses beyond it — exactly the predicted rule.

	m=0.005	m=0.01	m=0.02	m=0.05	m=0.1
$k = 20$ (k/n=0.005)	**2.83**	1.34	1.13	1.05	1.02
$k = 50$ (k/n=0.013)	**3.63**	**3.59**	1.52	1.14	1.06
$k = 200$ (k/n=0.050)	**3.70**	**3.69**	**3.67**	**3.52**	1.37
$k = 800$ (k/n=0.200)	**3.75**	**3.74**	**3.74**	**3.72**	**3.70**

most crowded documents

id	DTM radius	≈collisions at σ^*=0.12
`9b3dd305-0e50-4…`	0.921	≈12
`a310931d-78cc-4…`	0.922	≈12
`369ba45c-63b5-4…`	0.923	≈4
`7ce46beb-eb01-4…`	0.928	≈4
`18a8b3f6-a37c-4…`	0.933	≈3
`efaeae26-62ef-4…`	0.936	≈4

most isolated (voids)

`59c2a211-f8cc-4…`	1.207
`2a761d0e-c3e0-4…`	1.198
`0eba585f-60f9-4…`	1.190

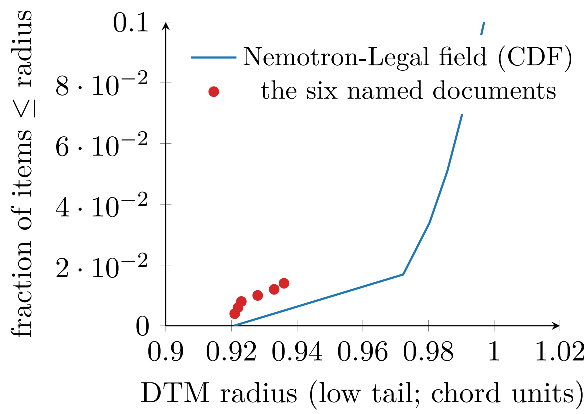

Figure 8: The per-entity localization readout, reproduced natively from the instrument's report on Nemotron-Legal (values are the report's own). *Left:* the crowded-document list — *named by uuid*, each with its DTM radius and its expected collision count at the corpus's σ^*: the radius says how crowded, the count says what it costs. *Right:* where those six sit — the extreme low tail of the field's CDF (the bulk rises to 1 near radius 1.05; the uniform band lies far right at 1.36). Aggregate, cell-based displays cannot provide this output.

a small perturbation of the corpus provably moves every item's score a small amount, the guarantee that step functions lack. The guarantee also holds empirically: jittering every coordinate by 10^{-3} leaves the field 0.997-correlated with itself. Against a fixed 10^6-row *synthetic* corpus — the benchmark generator scaled up, 5% pocket planted — the field's first percentile is invariant to the reservoir drawn (0.365–0.366 across reservoirs of 10^3–1.6×10^4 rows, ten seeds each), as is every other headline readout (σ^* spans 0.0559–0.0562 against 0.0564 at a 3.2×10^4 reference). On the traced benchmark the planted pocket drags the low tail to 0.364 against a bulk at 1.35 — the 3.7× separation behind the detection-power results of §8.

Both tails are actionable and *named*: the low tail is the crowded-item list (each entry annotated with its expected collision count at the corpus's own σ^*; §7), and the high tail is the list of coverage voids. In strongly cone-shaped corpora the entire field sits below the uniform band — the cone shortens every radius — and the instrument reports the regime explicitly rather than an uninformative all-items count; the finding in that regime is the shape of the low tail against the corpus's own bulk.

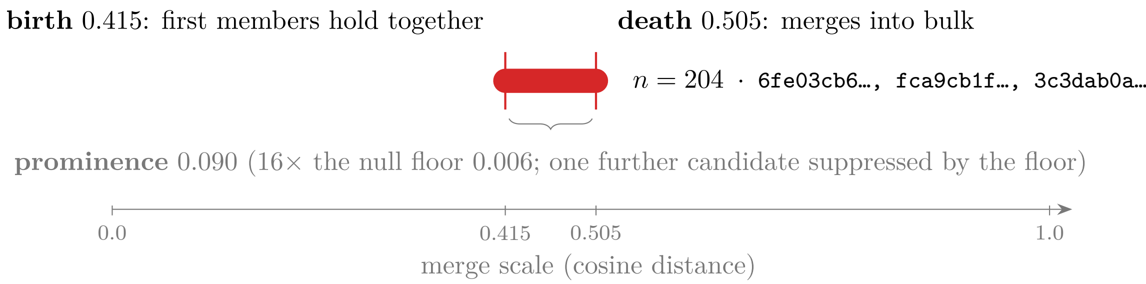

Figure 9: The pockets readout, reproduced natively from the instrument's report on Nemotron-Legal with the merge-tree anatomy annotated on it: the one pocket surviving the null-calibrated prominence floor — the largest prominence pure noise produces at matched size and dimension — (204 members, sample ids on the bar) is *born* when its first members hold together and *dies* into the bulk (excess-of-mass selection). Its prominence 0.090 is modest — a loose association, not a duplicate cluster — yet 16× the null floor, and therefore reportable; and the floor suppressed one further candidate; on the traced benchmark the same construction recovers a planted pocket exactly (200/200; birth 0.063, death 0.898, prominence 0.835), with no flat threshold anywhere.

6.4 Structure: the condensed merge tree

Between the global curve and the per-item field sits *structure*: one broad pile, or many tight pockets, and who constitutes each? `ambit` surfaces the hierarchy its clustering backend already computes and previously flattened: single linkage on the mutual-reachability metric (pair distance floored by each endpoint's distance to its own kth neighbor, which suppresses noise bridges) [35]. The tree is condensed at a minimum pocket size, and pockets are selected by excess of mass in $\lambda = 1/\text{distance}$: on each branch, the cluster persisting over the widest scale range is retained. This is equivalently H_0 persistence with the elder rule (at each merge, the younger component dies) [34, 36]. Each selected pocket reports birth, death, prominence (death minus birth: the scale range over which it persists as a separate entity), size, and member ids (Figure 9). Three implementation findings mattered — the third found by the validation battery itself.

- **Elder-rule inversion.** The naive elder rule is wrong for this purpose: the tightest pocket is the *eldest* component and survives to the root, so the bulk "dies into it." Excess-of-mass selection resolves the inversion.
- **Caterpillar membership.** Single-linkage chains inside a tight pocket are caterpillars (long chains with single-node side branches). Membership is therefore reported as the condensed core plus the side branches that fell out in the tight regime (below the geometric mean of birth and death), excluding bulk stragglers.
- **The childless root.** Chaining can prevent *any* real split from existing: a tight pocket may seed the one component that then absorbs the bulk point by point, so no node ever has two children of the minimum size, the only cluster is the root, and the tightest structure in the corpus goes unreported. A parameter sweep exposed this failure — at minimum size 32 a planted 200-member pocket returned nothing, and 3 of 20 benchmark

seeds missed it even at the default. The fix treats a childless root as a candidate with death at the final merge scale, whereupon the membership rule extracts exactly the pocket core.

Post-fix, the tree recovers the planted pocket in 20 of 20 seeds at exact size (200.0 ± 0.0; birth 0.063, death 0.898, prominence 0.835) across min-sizes 8–64, and on a two-pocket variant recovers sizes 60 and 40 exactly. Reported prominence also carries a *null floor*: uniform corpora of matched size and dimension produce maximum prominences of only ≈ 0.005 (99th percentile 0.009 over 400 nulls), so the instrument simulates the floor per run and suppresses pockets within it — a genuine pocket sits two orders of magnitude above it. Figure 9 shows the resulting readout on the real corpus: one pocket survives the floor (204 members, sample ids on its bar), and one further candidate was suppressed.

6.5 Complexity, and the design filters

All continuous-layer statistics run on the reservoir or the pair sample: the curve and its nulls are $O(P \log P)$ in the pair-sample size; the DTM field is a blocked Gram top-k, $O(n_r^2 d)$ on a capped subsample (seconds on CPU); the merge tree is a dense Prim MST on $n_r \leq 4{,}096$ rows plus a linear condensation pass. Full-corpus second moments stream in one pass.

Candidate metrics were admitted by five filters, recorded because they prune future proposals cheaply:

- *Spectrum:* does it reduce, under a linear kernel, to the covariance spectrum already reported?
- *Continuity:* is it a continuous functional of the data, or a step function of bin, rank, or partition membership?
- *Null:* does it come with a calibrated reference?
- *Naming:* can it score individual items, or only regions?
- *Supervision:* is it label-free?

These excluded, among others: RankMe, NESum, and stable rank (spectrum); k-means-conditioned statistics (continuity); raw counts (null); all cell aggregates (naming); and neural-collapse metrics [46] (supervision).

7 Operational Semantics: the Resolution Bandwidth

Geometric functionals quantify how uniformly a corpus covers the sphere; a practitioner needs to know how much query perturbation the corpus can absorb before retrieval degrades. This section converts the former quantity into the latter exactly, then measures the conversion.

7.1 An exact pairwise confusion law

Model a query aimed at item x as $q = x + \sigma g$, $g \sim \mathcal{N}(0, I_d)$ — the minimal model of a query that intends x but lands nearby. Retrieval errs on competitor y when $\langle q, y \rangle > \langle q, x \rangle$.

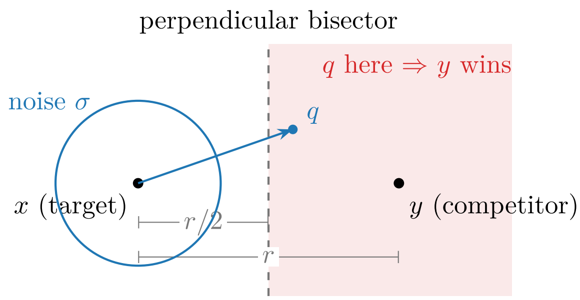

Figure 10: The confusion law's geometry: the decision boundary between target and competitor is the perpendicular bisector at distance $r/2$; Gaussian noise of scale σ crosses it with probability exactly $\Phi(-r/2\sigma)$. A far competitor is safe; a near-duplicate ($r \to 0$) has error probability approaching 1/2 at every noise scale.

Table 5: Verification of Proposition 2 by direct simulation at $d = 768$. Each cell uses 3×10^5 trials, enough to resolve deviations at the few-10^{-4} level of the Monte-Carlo error. Empirical flip rate vs. $\Phi(-r/2\sigma)$ with $r = 2\sin(\theta/2)$. Maximum absolute deviation over the grid: 9×10^{-4}, i.e. Monte-Carlo error — the law is exact, not asymptotic.

	$\sigma = 0.05$		$\sigma = 0.15$		$\sigma = 0.30$		$\sigma = 0.60$	
θ	emp.	pred.	emp.	pred.	emp.	pred.	emp.	pred.
0.1	.1583	.1588	.3700	.3695	.4335	.4338	.4677	.4668
0.3	.0014	.0014	.1592	.1596	.3101	.3092	.4021	.4017
0.6	.0000	.0000	.0245	.0244	.1626	.1623	.3113	.3112
1.0	.0000	.0000	.0007	.0007	.0554	.0550	.2114	.2121
1.4	.0000	.0000	.0000	.0000	.0156	.0159	.1416	.1415
2.0	.0000	.0000	.0000	.0000	.0024	.0025	.0803	.0804

Proposition 2 (Exact flip law)**.** *For unit vectors at chord distance* $r = \|x - y\|$, $\Pr[\langle q, y\rangle > \langle q, x\rangle] = \Phi\big(-r/2\sigma\big)$, *exactly, in any dimension.*

Proof. Geometrically (Figure 10): the set of queries for which y beats x is the half-space beyond the perpendicular bisector of x and y. The query starts at x, at distance $r/2$ from that boundary, and isotropic Gaussian noise carries it across with probability $\Phi(-(r/2)/\sigma)$. Algebraically: $\langle q, y - x\rangle$ is Gaussian with mean $\langle x, y - x\rangle = \cos\theta - 1 = -r^2/2$ and standard deviation $\sigma\|y - x\| = \sigma r$, so the flip probability is $\Phi\big((-r^2/2)/(\sigma r)\big) = \Phi(-r/2\sigma)$. □

Table 5 verifies the law across a 6×4 grid of angles and noise scales to a maximum deviation of 9×10^{-4}. Two readings follow: a competitor far relative to the noise is harmless, and a near-duplicate has error probability approaching 1/2 at every noise scale — the formal reason duplicate pockets are the worst crowding pathology (cf. §6.1's liftoff at the near-duplicate scale).

7.2 $C(\sigma)$, σ^*, and the uniformity bound

Summing the law over competitors gives the expected number of items outranking the intended target,

$$C(\sigma) = \sum_{j \neq i} \Phi\Big(-\frac{r_{ij}}{2\sigma}\Big) \;\approx\; (n-1)\int \Phi\Big(-\frac{r}{2\sigma}\Big)\, dK(r),$$

a functional of the pair-distance distribution alone. By the union bound (Boole's inequality), the sum bounds the probability of *any* error whatever the correlations among competitor events. C is therefore conservative: hubness — a few items recurring as near neighbors of many others, which correlates those events — makes the bound loose, never wrong.

Definition 3 (Resolution bandwidth)**.** *σ^* is the largest σ with $C(\sigma) \leq 1$: the most noise queries can carry before, in expectation, one wrong item outranks the right one. C is monotone; bisection inverts it.*

Figure 11 shows the real corpus's curve against its uniform control: Nemotron-Legal crosses the one-collision tolerance at $\sigma^* = 0.123$ against 0.148 for a well-spread corpus of the same size and dimension — it retains 83% of the ideal noise budget — while on the traced benchmark the planted 5% pocket spends 26% of the budget ($\sigma^* : 0.203 \to 0.150$). The per-coordinate scale σ has a direct reading in cosine units. A noisy query's expected similarity to its own target decays as $\langle \bar{q}, x\rangle \approx (1 + \sigma^2 d)^{-1/2}$, so σ^* states how dissimilar a query may become from its intended target and still be expected to win. At $d = 1024$, $\sigma^* = 0.123$ corresponds to a query decayed to cosine ≈ 0.25 against its target. The lower panel of Figure 11 overlays the five corpora under the same encoder; the spread of their tolerance crossings, $\sigma^* = 0.090$ to 0.129, is the corpus-to-corpus noise-budget comparison the instrument is designed to support.

Proposition 3 (Uniformity as a Chernoff bound)**.** *With $U_t = \log \mathbb{E}\, e^{-t\|x-y\|^2}$ the uniformity functional of [7]: $C(\sigma) \leq (n-1)\, e^{U_t}$ at $t = 1/(8\sigma^2)$.*

Proof. $\Phi(-z) \leq e^{-z^2/2}$ with $z = r/2\sigma$. □

The widely used uniformity temperature is thus not arbitrary: $t = 2$ is a collision bound at query noise $\sigma = 0.25$. `ambit` reports uniformity with this reading attached.

7.3 Scope

The guarantee has three boundaries.

- *Query population.* The corpus is treated as its own query population. This is exact for deduplication, clustering, and corpus-as-queries retrieval; extrapolating to an external workload assumes the workload lands where the documents are.
- *Noise shape.* The noise channel is isotropic, while real query drift is structured. The sensitivity of this assumption is measured. Re-ranking a grid of corpus types under structured Gaussian channels at matched total power, the isotropic ordering survives moderately structured noise (rank-25 subspace noise: 0.99 pairwise-order agreement over five seeds). It degrades under extreme misspecification (rank-2: 0.79; noise aligned with each corpus's own top principal directions: 0.82). Comparative use is therefore robust; adversarial noise structure is the stated failure mode.

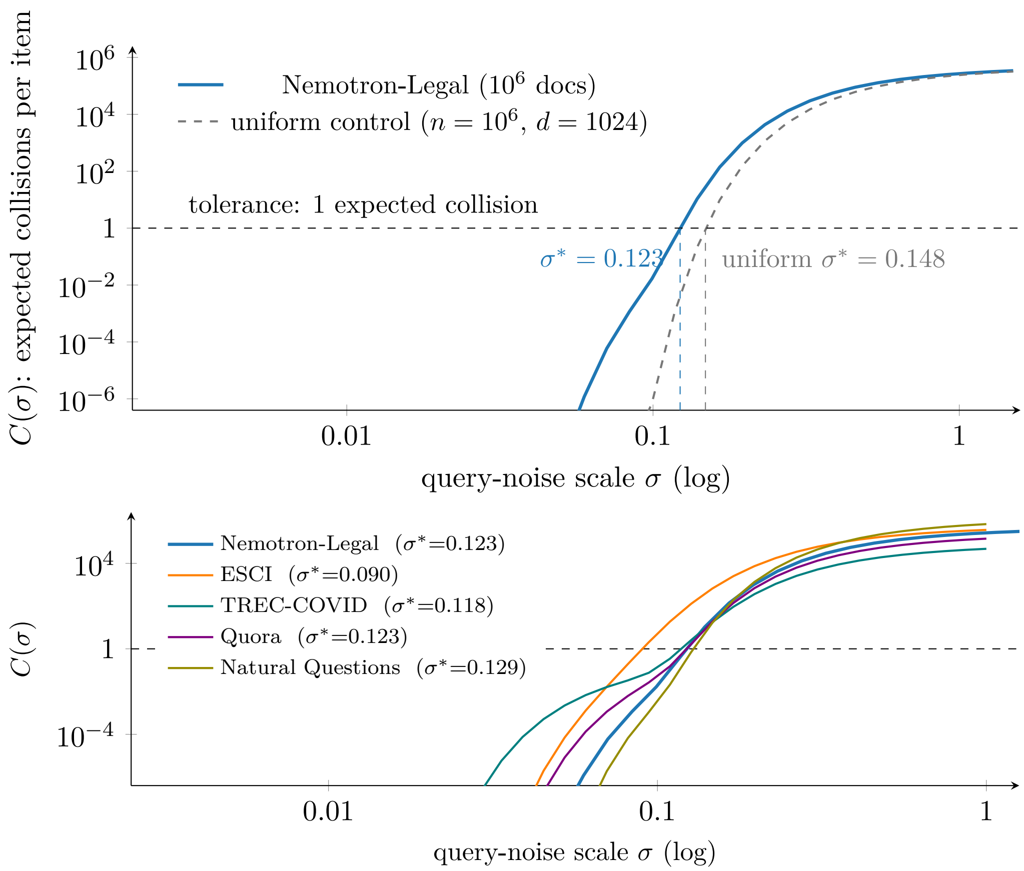

Figure 11: Expected collisions vs. query noise on the real corpus (measured; Nemotron-Legal). Top: the corpus against its uniform control — the corpus's cone and tight pairs dominate $C(\sigma)$ at the noise scales where retrieval operates, pulling the tolerance crossing from $\sigma^* = 0.148$ to 0.123. At large σ the curves converge (every pair confuses) and at tiny σ both vanish; the tolerance crossing is determined where they differ. Bottom: the five corpora under the same encoder, with tolerance crossings spanning $\sigma^* = 0.090$ to 0.129; the text gives the reading.

- *Aggregation.* The aggregate is an upper bound: C over-counts confusions, never under-counts them.

σ^* is therefore a conservative intra-corpus guarantee, strongest used comparatively — one corpus embedded two ways, or one embedding over time.

7.4 Negative results

Two negative results from the design process are recorded to prevent their re-derivation.

Near-redundancy of C with uniformity (Table 6). As a corpus-level *discriminative* measurement, the confusion transform adds almost nothing beyond uniformity: rank correlation 0.978, with 27 of 28 pairwise orderings in agreement. Its value is semantic instead — units, calibration, and per-item counts. The single disagreement is informative. The confusion kernel ranks a duplicate-pocket corpus as worse than a diffuse low-rank collapse; the Gaus-

Table 6: Measured near-redundancy of the confusion functional with uniformity across eight synthetic corpus types (n = 2,000, d = 256; C at $\sigma = 0.25$, U at the matched $t = 2$). Rank correlation 0.978; 27 of 28 pairwise orderings agree. The one disagreement (boldface) is between qualitatively different pathologies — and the confusion ordering is the operationally correct one.

corpus type	$C(0.25)$	$U_{t=2}$
uniform	4.86	−3.97
one cluster of 50	4.90	−3.96
twenty clusters of 40	5.41	−3.90
one clump of 200	5.58	−3.87
five clumps of 100	5.80	−3.84
anisotropic cone	8.38	−3.48
low-rank collapse	**15.49**	**−3.08**
cone + duplicate pocket	**17.78**	**−3.21**

sian kernel ranks them oppositely. Duplicate pairs approach error probability 1/2 at every noise scale, while a spread subspace remains partially navigable, so the confusion ordering is the operationally correct one. The flip occurs precisely between qualitatively different pathologies.

A vacuous stability certificate. Composing the DTM stability bound with a split-half empirical Wasserstein estimate, hoping for finite-sample error bars, fails structurally at embedding dimension: the matching distance between independent halves is of the order of the data scale (≈ 1.35 on $\mathbb{S}^{767}$ at $n_r = 2{,}000$), so the resulting certificate is $m^{-1/2}\widehat{W}_2 \approx 13.5$ — an order of magnitude beyond the field's entire range. Dimension-free certificates would need a different metric with its own stability theorem — an open question, not an engineering task.

8 Calibration and Detection Power

The instrument's statistics were calibrated and stress-tested as a whole against 400 pure-null corpora: corpora drawn with no planted pathology, so that any alarm is a false alarm. Four hundred nulls estimate a 1% false-alarm rate to a standard error near 0.5%. The instrument was then compared for detection power against a suite of standard diagnostics, with *every* threshold — the instrument's and the baselines' — set on those same nulls at a matched 1% false-alarm rate, so that all detectors are compared at fixed specificity.

Null calibration. On pure nulls (n = 2,000, d = 256; the benchmark family's generators with no pathology): the global rank test rejects at $p \leq 0.01$ in 1.25% of nulls (nominal 1%); the gated liftoff fires in 6% at $\alpha = 0.05$ (nominal 5%); $|z| > 3$ occurs in 0.75%; and the pocket-prominence maximum has null median 0.005 and 99th percentile 0.009 — the floor of §6.4. Measured false-alarm rates for all calibrated thresholds land on 0.010–0.013.

Detection power. Table 7 reports the pathology grid. Three observations follow. First, the continuous layer — the rank test, σ^*, the DTM tail ratio, and pocket prominence — is at *full power in every pocket cell.* That includes the hardest cell, a single pocket of 20 items

Table 7: Detection power at matched 1% false alarm (50 replicates per cell, enough to estimate each power entry to a standard error of at most 0.07; thresholds calibrated on 400 nulls; $n = 2{,}000$, $d = 256$). Pocket cells are (count × size, spread s; $s = 0.005$ is near-duplicate, 0.06 loose). Continuous-layer detectors left of the rule; baseline suite right. The hardest cells — a single pocket of 1% of the corpus — separate the layers; boldface marks the exceptions discussed in the text.

cell	rank_p	σ^*	DTM	pocket	meancos	Iso	unif.	hub	kNN_{q05}
1×20, s=.005	1.00	1.00	1.00	1.00	.04	**1.00**	.88	.18	.22
1×20, s=.02	1.00	1.00	1.00	1.00	.02	**1.00**	.70	.42	.34
1×20, s=.06	1.00	.22	1.00	1.00	.04	**1.00**	.02	.90	.32
1×50, s=.005	1.00	1.00	1.00	1.00	.58	1.00	1.00	1.00	.92
5×20, s=.02	1.00	1.00	1.00	1.00	.26	1.00	1.00	1.00	1.00
1×200, any s	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
cone (mean shift)	1.00	1.00	1.00	**.12**	1.00	**.00**	1.00	1.00	1.00
low-rank collapse	1.00	1.00	1.00	1.00	.26	1.00	1.00	**.00**	1.00

holding 1% of the corpus. On that same cell, mean-cosine (0.04), hubness (0.18), and the kNN-distance quantile (0.22) sit near the false-alarm floor: too little of the corpus participates in the pocket to move any average. This quantifies the claim of §2.3. Second, the exceptions are reported explicitly. Covariance-spectrum statistics (IsoScore) are *also* at full power on single coherent pockets: geometrically, a 1% mass sharing one direction lifts an eigenvalue past the null spectrum edge, so for pure corpus-level detection of one tight pocket a spectrum summary suffices. What a spectrum summary cannot do is name the pocket's members, detect a mean-shift cone (power 0.00 there, because centering removes the shift), or express the loss in operational units. Third, the discriminating cells behave as the theory says they should: the mean-chord z is weak on small pockets (their pair mass is tiny); the pocket detector correctly declines to alarm on a cone (0.12 — a cone is not a pocket); and hubness is blind to a low-rank collapse. The layer's value is therefore not a single more powerful scalar but full power at matched specificity *together with* localization to named items, scale attribution, and operational units — the properties the baselines structurally lack.

9 The Instrument

9.1 Architecture

`ambit` is a library with a thin CLI. One canonical in-memory type (`EmbeddingSet`) flows through a pipeline: a *streaming scan* (one pass over `.npy/.npz/.parquet/.csv/.jsonl`, single file or sharded directory) accumulates exact second moments, norms, and a uniform reservoir; a context builder derives projections (PCA by default), the eigenspectrum, a kNN graph (optionally filtered to *reciprocal* neighbors, which suppresses hubs by construction), unsupervised clusters, and the pair-cosine sample. All measurements of §6–§7 are library functions over these objects; a separate presentation layer (Appendix Appendix B) renders them. The core depends on `numpy` alone; heavier capabilities (tabular IO, ANN graphs, UMAP, GPU paths, on-the-fly embedding via an OpenAI-compatible endpoint, the training extra) are optional and degrade gracefully. A comparison mode aligns two embeddings of the same

Table 8: Scaling on the measurement host (numpy-only core; raw `.npy` input; per-phase wall time and peak RSS from forked children — RSS is dominated by memory-mapped input pages, not allocator footprint). Left: corpus-size sweep at $d = 1024$; the streaming scan is linear in bytes at $\approx 2\,$GB/s and the reservoir-bound phases are *constant* in n. Right: dimension sweep at $n = 10^6$; the scan carries the d^2 scatter accumulation and the context build the d^3 eigendecomposition, while the continuous layer stays nearly flat.

$d = 1024$			
n	scan	ctx	cont.
10^5	0.2 s	2.7 s	1.8 s
10^6	2.0 s	3.0 s	1.8 s
$3{\times}10^6$	5.6 s	3.0 s	1.8 s
10^7	18.6 s	2.6 s	1.8 s

$n = 10^6$			
d	scan	ctx	cont.
256	0.6 s	0.5 s	1.8 s
1024	2.0 s	3.2 s	1.8 s
2048	4.6 s	10.4 s	2.0 s
4096	12.4 s	52.6 s	3.4 s

items by id and computes a local neighbor-overlap view read against global representational-similarity statistics (centered kernel alignment, CKA [45], kernel and rigid distances); the local/global pair can disagree informatively — a high-CKA, low-overlap outcome indicates an embedding change that preserves gross structure while rewriting fine neighborhoods.

9.2 Scaling

Table 8 gives the measured curves. The pipeline behaves as designed: the one pass over the corpus is I/O-bound and linear in bytes (a $10^7 \times 1024$, 41 GB corpus streams in 18.6 s on raw `.npy`; the Parquet path adds decode overhead — the case study's $10^6 \times 1024$ sharded-Parquet run takes $\approx$ 3.5 minutes end to end on the smaller host), while everything downstream of the scan is *reservoir-bound and therefore constant in corpus size*: context build $\approx 3\,$s and the entire continuous layer $\approx 1.8\,$s whether the corpus has 10^5 or 10^7 rows. Dimension enters through the scan's d^2 scatter accumulation and the context build's d^3 eigendecomposition (52.6 s at $d = 4096$ — the one super-linear phase), while the continuous layer grows only through BLAS on the reservoir Gram ($1.8 \to 3.4\,$s over $d = 128 \to 4096$).

One caveat is demonstrated empirically. With the planted pocket placed in the corpus's *last* rows (the layout produced by label-sorted shards), the full scan detects it (rank test $p = 0.010$, liftoff 0.92), while approximate scans of the first $5{\times}10^4$ to $3{\times}10^5$ rows see a clean corpus ($p = 0.18$–0.45, no liftoff, mean cosine ≈ 0): an approximate scan's reservoir is only as representative as the row order that fed it, so the instrument's approximate mode is for exploratory iteration rather than for conclusions on data whose layout may correlate with content.

9.3 Case study: a million-document corpus

Table 9 summarizes a full run on the Nemotron-Pretraining-Legal-v1 corpus [60] (§4): 10^6 documents from four legal subsets, embedded at $d = 1024$ by an instruction-tuned text encoder. The corpus is strongly cone-shaped (mean pair cosine +0.236; IsoScore 0.090) yet spectrally rich (effective rank 728/1024); the occupancy discrepancy sits at $z = -4{,}211$, and the whole-curve rank test rejects uniformity at the minimum attainable $p = 0.005$ with liftoff at cos ≈ 0.82 — genuine tight-pair excess, at the near-duplicate scale. One correction

Table 9: Case study: Nemotron-Pretraining-Legal-v1 [60] (four subsets, 10^6 documents), $d = 1024$; full report in $\sim$3.5 minutes on CPU, one streaming pass.

items $\times$ dims	1,000,000 $\times$ 1024
mean pair cosine	+0.236
IsoScore / effective rank	0.090 / 728.0/1024
occupancy discrepancy z	$-4{,}211$
whole-curve rank test	$p = 0.005$; liftoff cos ≈ 0.82
resolution bandwidth σ^* (uniform null)	0.123 (0.148)
noise budget retained	83%
top pocket (size; birth; prominence)	204; 0.41; 0.09
kNN purity / silhouette (4 provided subsets)	0.95 / +0.04
most-crowded items	$\approx$ 3–12 expected collisions each

is attributable to the calibration study: an earlier pointwise reading placed the liftoff at cos $\approx +0.07$; that reading was precisely the bulk-edge multiple-comparison artifact that the null calibration exposed; the corrected test relocates the significant excess to the high-similarity scale. The bulk of the discrepancy z remains the cone, per the ACG reference. Operationally the corpus retains 83% of the ideal noise budget: a query may fade to cosine ≈ 0.25 against its target before an impostor is expected to win. The per-item field's low tail is real but shallow, and the merge tree finds one modest pocket: crowding here is *diffuse within subsets* rather than pocketed — the regime where global averages mislead optimistically while per-item counts still name the at-risk documents. The separability panel (labels provided, used only to condition the readout) shows high neighborhood purity (0.95) against near-zero silhouette (+0.04): subsets are locally coherent yet globally entangled within the cone — a reading no single number provides.

10 From Measurement to Gradient

The preceding sections establish that the instrument detects, localizes, and prices crowding. This section and the next test a stronger claim: that the measurements suffice to *drive* interventions. Four interventions are tested:

1. a measurement-derived training objective (below);
2. a measurement-gated fine-tuning loop on the case-study encoder (§10.3);
3. execution of the instrument's deduplication prescription (§10.5);
4. measurement-derived guardrails for supervised fine-tuning (§10.6).

The confusion kernel is differentiable, so the measurement layer doubles as a training-time regularizer with a stated guarantee.

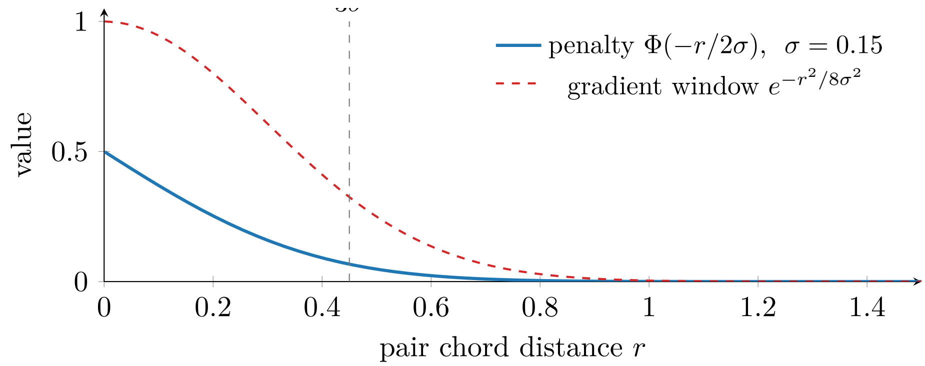

Figure 12: The confusion loss's gradient locality at a measured $\sigma = 0.15$ (the benchmark's σ^*): all gradient lives inside the confusable window; pairs beyond $\approx 3\sigma$ are exponentially untouched. "Improve resolution without disturbing dissimilarity" is guaranteed by the kernel.

10.1 Losses and the gradient-locality guarantee

For a batch $z_1, \ldots, z_B$ (normalized) and a *measured* target scale σ:

$$\mathcal{L}_{\text{conf}} = \operatorname*{mean}_{(i,j)\notin\mathcal{E}} \Phi\left(-\frac{\|z_i - z_j\|}{2\sigma}\right), \qquad \mathcal{L}_{\text{pres}} = \operatorname*{mean}_{i} \ \mathrm{KL}\left(p_i^{\text{base}} \,\middle\|\, p_i\right),$$

where $\mathcal{E}$ excludes positive and guarded pairs and p_i (p_i^{base}) are temperature-softmax similarity rows under the tuned (frozen base) model. $\mathcal{L}_{\text{conf}}$ is the batch estimate of $C(\sigma)$; since $\partial\Phi(-r/2\sigma)/\partial r \propto e^{-r^2/8\sigma^2}$ (Figure 12), its gradient vanishes exponentially for pairs beyond $\approx 3\sigma$: far-pair (dissimilarity) structure *cannot* be disturbed, a property enforced in the test suite (a near pair receives $> 10^3\times$ the gradient of an orthogonal pair). $\mathcal{L}_{\text{pres}}$ is the neighbor-overlap comparison in differentiable form — the explicit statement of "do not rewrite who is similar to whom." (Implementation note: the chord computation floors $2 - 2c$ before the square root; at exactly zero the root's infinite derivative poisons masked entries with $0 \cdot \infty$ NaN gradients.)

10.2 Mining, protocol, and measured effect

Batch composition is driven by the measurements in three ways.

- *Sampling:* items are drawn with weight proportional to their per-item collision counts, with a uniform floor so that the uncrowded bulk still anchors the geometry.
- *Negatives:* candidate negatives are drafted from the *confusable window* — the cosine range between the measured liftoff, where confusability begins, and the duplicate scale, where identity begins (operationalized as $\cos \approx 0.98$ throughout).
- *False-negative guard:* an anchor's top-m base-model neighbors are never used as negatives, because the confusable window is precisely where unlabeled true relatives live.

The protocol is staged.

Table 10: The worked training example (a planted 300-item pocket in 4,000 items at $d = 64$; a linear adapter as the model stand-in; all verdicts from a held-out slice unseen by mining or batching). A linear adapter roughly halves flagged-item collisions at high structure preservation; we report this as the attainable ceiling of the *adapter* — which cannot unfold a tight pocket further without moving everything else, exactly the trade $\mathcal{L}_{\text{pres}}$ enforces — not of the method.

schedule	flagged median collisions	neighbor overlap w/ base
before	2.59	1.00
Adam, 10^{-2}, 600 steps, λ_p=0.3	1.25	0.92
Adam, 3×10^{-2}, 600 steps, λ_p=0.3	1.32	0.91
Adam, 3×10^{-2}, 1000 steps, λ_p=0.5	1.60	0.93

1. *Diagnose*, and take the cheapest fix that the readouts license: a cone licenses linear post-processing; duplicates license data repair; only genuine beyond-cone clustering licenses training.
2. *Train* with $\mathcal{L}_{\text{task}} + \lambda_c\mathcal{L}_{\text{conf}} + \lambda_p\mathcal{L}_{\text{pres}}$ at the measured σ.
3. *Verify* on a held-out reservoir never seen by mining, and never grade with the functional that was optimized.

Table 10 reports the worked end-to-end example. Flagged items — the instrument's crowded-item list, the highest expected collision counts under the base model — see their median expected collisions fall from 2.6 to 1.25 at held-out neighbor overlap 0.92; raising the preservation weight trades recovery for overlap, exactly as designed. The guard matters empirically: without it, window mining drafts unlabeled true relatives as negatives — the classic silent failure of hard-negative mining.

10.3 Closing the loop on a full-scale encoder

The synthetic example establishes the mechanism; the remaining question is whether the loop — measure, construct the objective from the measurements, train, re-embed, re-measure — operates on a contemporary public encoder and a million-document corpus, with the measurements adjudicating every round. We ran this loop on the case-study corpus (§9) and its 0.6-billion-parameter instruction-tuned encoder. A fixed, stratified 200,000-document measurement subset (proportional across the four subsets) was drawn once; 20% was held out and never touched by mining, sampling weights, or training. The objective is fully unsupervised: $\mathcal{L}_{\text{conf}}$ at the subset's measured $\sigma^* = 0.134$ under the false-negative guard, plus $\lambda_p\mathcal{L}_{\text{pres}}$ against frozen base-model references, with batches composed half from collision-weighted sampling and half from guarded confusable-window pairs (the confusable window of §10: liftoff 0.769 to 0.98). Between rounds, a pre-registered licensing rule decides continuation: σ^* must not fall and held-out neighbor overlap must remain at least 0.90 — that is, tuning may rewrite at most one held-out neighbor in ten.

Figure 13 plots the loop's held-out verdicts per round.

Table 11 summarizes the rounds; three findings follow. First, *the instrument rejects a misspecified round*. The initial LoRA round computed $\mathcal{L}_{\text{pres}}$ against stored embeddings

Table 11: The loop on the case-study corpus (200k measurement subset; verdicts from the held-out 20% only; flagged cohort = top 1% of items by base-model expected collisions, tracked at fixed $\sigma = 0.134$ across rounds). The misspecified first round is *rejected by measurement*; the corrected rounds are safe but marginal at every capacity.

round	σ^*	liftoff	z	pocket (size)	overlap	cohort med.
base	0.1337	0.769	−3565	0.085 (294)	1.00	2.58
linear adapter	0.1331	0.770	−3665	0.064 (17)	0.979	2.82
LoRA, mismatched refs	0.1324	0.758	−3775	0.091 (319)	0.924	3.01
LoRA, corrected	0.1339	0.763	−3545	0.082 (307)	0.973	2.51
full fine-tune	0.1338	0.763	−3552	0.083 (275)	0.965	2.53

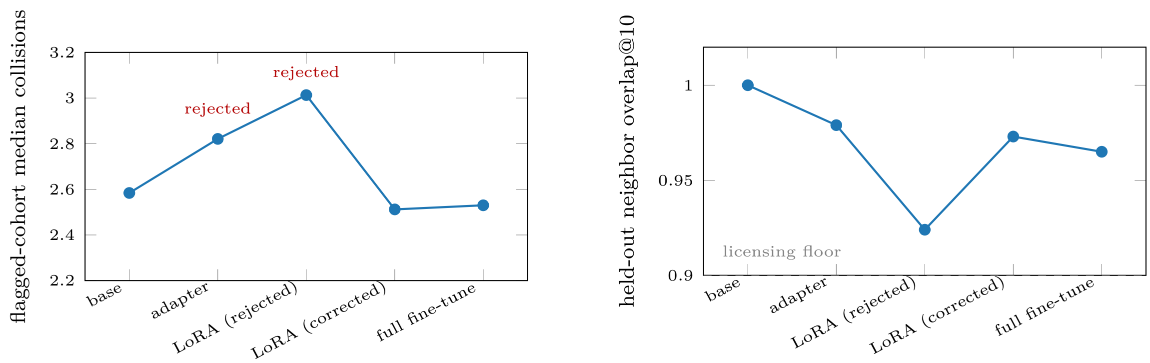

Figure 13: The encoder loop's held-out verdicts by round (values from Table 11). Left: median expected collisions of the flagged cohort, tracked at fixed σ — the two rejected rounds *worsen* the cohort while their training losses decreased; the corrected rounds improve it. Right: neighbor overlap against the base model, with the pre-registered licensing floor; every accepted round stays above it.

of full documents while the trainer embedded token-truncated views, so the gradient was dominated by a length distillation (the model learning to map truncated views onto full-document embeddings); training losses fell while every global readout degraded (σ^* down, cohort collisions up 17%, the pocket enlarged). The licensing rule caught this from the held-out measurements alone. Second, with references matched to the trainer's own views and the confusable window placed in every batch, both LoRA and full fine-tuning are *safe but marginal*: no readout degrades, held-out overlap stays at 0.97, and the flagged cohort improves by 2–3% in the median. The effect is small but nearly universal across items: of the 408 flagged items that fall in the held-out split, 406 (LoRA) and 356 (full fine-tune) individually reduce their expected collisions (paired sign tests, $p < 10^{-100}$ and $p < 10^{-56}$). σ^* is measured as genuinely unchanged rather than noisily so: the paired difference across eight independent pair samples is $+0.00007$ against a per-sample standard deviation of 0.00012. All of this despite the full fine-tune moving roughly $40\times$ the parameters of the LoRA round on a doubled pair batch. Third, the capacity ladder — linear adapter, LoRA, full fine-tune — shows that *capacity is not the binding constraint*. The linear adapter, which halved collisions on the synthetic single-pocket benchmark (Table 10), is itself rejected on the real corpus: σ^*

falls and cohort collisions worsen by 9%, because a single linear map cannot serve the corpus's mixture of scales. LoRA and the full fine-tune land within noise of one another. The corpus's dominant pocket is textual near-duplication, and the preservation term correctly refuses to separate items whose inputs are near-identical. This is the training-side confirmation of the staged protocol's routing rule: duplicate pathology is a data-repair problem, and no training objective that preserves base-model semantics can substitute for deduplication. Training at the measured scale is safe; it is not a shortcut around the data.

As a final check that the subset verdicts transfer, the full corpus was re-embedded with the accepted fine-tune and measured at full scale with the identical pipeline as the base case study. Every readout reproduces within measurement noise — σ^* 0.1227 against the base 0.1226, liftoff 0.817, effective rank 728, IsoScore 0.089, mean pair cosine 0.235, hubness skew 1.88, top pocket 198 items at prominence 0.090 against the base 204 at 0.090 — so the fine-tune that improves retrieval (below) is confirmed geometry-preserving at 10^6 documents, and the 200k held-out measurement is a faithful proxy for the full corpus.

Reading Figure 13. The left panel is the counterfactual that the instrument prevented: both rejected rounds sit *above* the baseline — the tuned models made the flagged cohort worse — while their own training losses were falling monotonically. Training loss alone endorsed that round; only the held-out geometric readouts identified the regression. The right panel shows the gate that caught it: the rejected LoRA round is the only point below the pre-registered overlap floor, and the accepted rounds cluster near the baseline's neighborhood structure. The instrument's contribution here is not the (marginal) improvement of the accepted rounds; it is that on every round that the labels later scored, the held-out geometric verdict agreed in sign with the label-based evaluation, at zero label cost.

10.4 Driving retrieval behavior

The loop above never touches a relevance judgment; judgments enter exactly once, after all rounds are frozen, to *score* the frozen predictions (the blind-then-score discipline of the validation plan). On the corpus's citation-grounded question-to-section retrieval set (1,000 held-out questions against a 35,173-section sub-corpus, exact search), the unsupervised full fine-tune moves recall@1 from 0.288 to 0.301 and MRR@10 from 0.467 to 0.475; the LoRA round is flat (0.286, 0.466). The recall gain does not reach significance at this query count: McNemar's exact test on the paired outcomes gives $p = 0.45$ (118 against 131 discordant queries), and the bootstrap interval on the MRR@10 difference, $[-0.008, +0.023]$, spans zero. The correct reading is therefore *retrieval-neutral-to-positive*: a per-item-significant geometry improvement whose aggregate task effect is bounded near zero, consistent with the duplication finding — most failures are not caused by the crowding the objective can legally repair.

For context, a label-supervised fine-tune measured under the same harness reaches recall@1 0.355, an upper reference point that supervision, not geometry, supplies.

The per-item layer makes a sharper, item-level prediction: documents with high base-model expected collisions should be the ones that *fail at their own queries* (a judged-relevant document outranked by non-relevant neighbors). Scored blind, base expected-collision counts predict failure participation at AUC 0.578 against 0.551 for the natural control (top-1 neighbor cosine) over 953 gold documents. The signal is significantly better than chance (bootstrap 95% interval $[0.536, 0.622]$) but the margin over the proximity control is not individually significant on this single corpus (+0.028, paired bootstrap interval $[-0.017, +0.073]$) — establishing

Table 12: Dedup routing scored blind on 1,000 frozen queries (fixed/broken = queries whose failure status at rank 10 changed). Confusable-window cleanup applies to foreign distractors; only true-duplicate merging applies within the formulaic corpus.

threshold	foreign distractors removed	target corpus merged
liftoff (0.82)	$+12/-0$ fixed, $p = 0.0005$	$+26/-57$, $p = 0.0009$ (harmful)
duplicates (0.95)	0/0 (no effect)	$+9/-2$, $p = 0.065$

the cross-corpus version of this comparison is a designated hypothesis of the preregistered external-validation plan.

In operational terms the instrument drives retrieval behavior through the readout-to-intervention map exercised end to end in this section: the anisotropy layer licenses linear post-processing; named near-duplicate pockets route to data repair before any training; genuine beyond-cone clumping licenses guarded training at the measured σ^*, verified on held-out measurements with a pre-registered continuation rule; per-item collision counts triage which documents to repair or re-chunk first; and σ^* itself sets the query-noise budget an application may spend.

10.5 Executing the routing rule

We tested the routing claim directly, executing the repair that the measurement prescribed while blind to all labels, then scoring on the frozen evaluation. The repair has three mechanical steps:

1. identify near-duplicate groups purely from geometry (pairs above a stated cosine threshold, grouped by connectivity);
2. reduce each group to one canonical member, chosen by identifier order;
3. map relevance judgments through that reduction mechanically.

Two repair sites and two thresholds give a double dissociation (Table 12): each repair helps at exactly one site-and-scale combination and harms or does nothing at the others. Removing *foreign* near-duplicates inside the confusable window (the liftoff scale) fixes 12 queries and breaks none (McNemar exact $p = 5 \times 10^{-4}$); this is the significant retrieval gain that training could not deliver. Merging *within* the formulaic target corpus at that same scale is significantly harmful (57 queries broken): these regulatory sections share boilerplate structure, so their confusable window contains legally distinct sections. Only merging at the true-duplicate scale ($\cos \geq 0.95$) helps within the corpus. The instrument's scale structure — liftoff marks the onset of confusability, not the duplicate boundary — prescribes different repairs at different scales, and the held-out labels confirm each mapping in both directions (Figure 14).

Reading Figure 14. "Deduplicate the corpus" names a single operation, but its effect depends on site and scale: of the four cells in the figure, only one is cleanly positive. The measurement supplied the two numbers that disambiguate the operation — the liftoff cosine, where confusion begins, and the duplicate scale, where identity begins — and each licenses a different intervention.

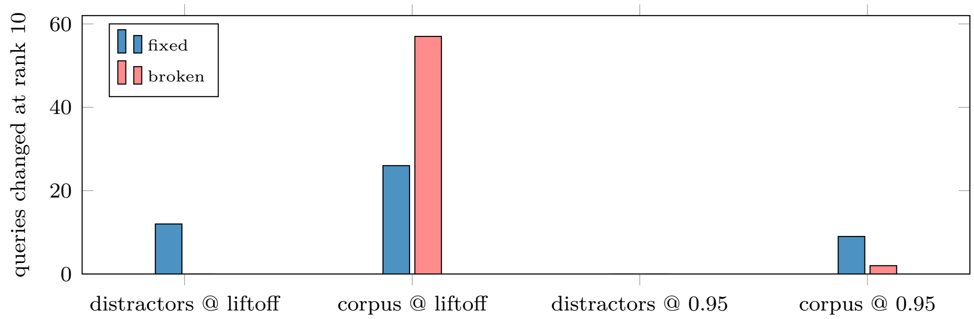

Figure 14: The routing dissociation as counts (values from Table 12). At the liftoff scale, removing *foreign* confusable material only fixes; merging *within* the formulaic corpus at that same scale mostly breaks. At the true-duplicate scale the pattern reverses in both sites. Each repair is licensed at exactly one scale and one site — the mapping the instrument's readouts prescribe.

10.6 Guardrails for supervised fine-tuning

The strongest test of the training layer reverses its role: rather than driving an unsupervised objective, the measurements *guard* a label-supervised one. A prior supervised fine-tune of the case-study encoder had gained first-rank retrieval while silently collapsing an untargeted multi-hop evaluation. This is exactly the failure class that $\mathcal{L}_{\text{pres}}$ and the false-negative guard exist to prevent. We ran the identical supervised recipe (in-batch contrastive loss on 45,465 question–section pairs, two epochs) in two arms differing only in the guardrails.

- *Control:* the task loss alone.
- *Guarded:* the task loss plus $\lambda_p \mathcal{L}_{\text{pres}}$ against the frozen base on identical truncated views. In addition, any batch document whose base-model cosine to the anchor's positive exceeds the measured liftoff is masked out of the contrastive denominator, so the task loss never pushes apart pairs that the base model considers confusable relatives.

The held-out measurement gate, read before any judgment, ranked the arms: the guarded model dominates on every label-free readout (neighbor overlap 0.650 against 0.537; σ^* 0.146 against 0.136, the base being 0.134; flagged-cohort collisions 0.47 against 2.10). The frozen evaluations then confirmed the prediction. On the targeted task the guardrails cost nothing (MRR@10 0.503 against 0.501; both arms far above the base 0.467). On the untargeted open evaluation the guarded arm wins everywhere. Single-document MRR@10 is 0.597 against 0.563. The decisive metric is all-gold@10 — the fraction of multi-hop queries whose every judged-relevant document reaches the top ten — because that is where the historical regression lived. There the guarded arm scores 0.339 against the control's 0.223, a 52% relative improvement over 1,496 held-out queries; the gap exceeds five standard errors. The control reproduces the historical collapse; the guard halves the damage while improving every other number. Two conclusions follow: the measurement-derived guardrails make supervised domain adaptation safer at no cost to its gains, and the instrument's label-free held-out gate selects the better model before any label is consulted. Figure 15 presents the comparison.

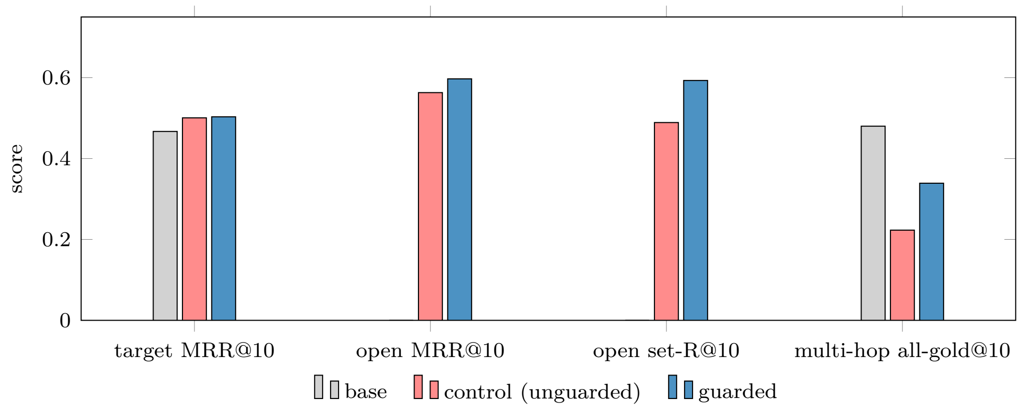

Figure 15: The guardrail experiment on the frozen evaluations. On the target task (leftmost group) the guarded arm matches the control — the guardrails are free. On the untargeted open evaluation (set-R@10 is the recall of the full judged set within the top ten) the guarded arm wins every metric, most sharply on the multi-hop all-gold@10, where the control reproduces the historical regression (base ≈ 0.48, historical baseline; base bars for the open single/set metrics were not measured under this harness and are omitted).

Reading Figure 15. Compare the control and guarded bars group by group, left to right. On the targeted metric the two are indistinguishable: whatever the guard rails removed from the training signal, none of it was needed for the supervised gain. The three open-eval groups then separate progressively, widest on the multi-hop metric — exactly the ordering one predicts if the control's damage is concentrated in *relations among documents* rather than query–document matching, because the guard's masked pairs and the preservation term are both relational objects. The mechanism is visible in the label-free gate that preceded scoring: the control had reshuffled 46% of held-out neighborhoods (overlap 0.537), the guarded arm 35% — and neighbor identity is what multi-hop retrieval consumes.

11 External Validation Across Corpora and Encoders

The preceding sections validate the instrument internally and on one corpus. This section reports a first cross-corpus, cross-encoder slice of the external-validation design: the four judged collections and seven encoders of Tables 1 and 2. Each combination of a corpus and an encoder — a *cell* of the grid — is treated as one embedded corpus to be measured and then scored, giving 28 cells plus the legal case study. Each cell follows four steps in strict order:

1. embed the corpus;
2. compute the full readout set and record its hash, *before* any judgment file is opened;
3. compute the per-document collision field of §7 (each document's expected collision count) at the cell's measured σ^*;
4. only then run and score the queries (nDCG@10, recall@100, and per-relevant-document failure labels).

Table 13: The external grid at a glance (7 encoders per corpus; readouts frozen before judgments). ρ is the within-corpus Spearman rank correlation across encoders between the readout and nDCG@10; "control" is the same correlation for mean pair cosine. The final columns give the range, across encoders, of the per-document collision field's failure-prediction AUC (a judged-relevant document ranked past 10 at its own queries) and the count of encoders for which that AUC exceeds chance.

corpus	task type	$\rho(\sigma^*)$	control	failure AUC	above chance
Quora	duplicate matching	+0.75	−0.79	0.66–0.74	7 of 7
ESCI	product search	−0.07	+0.32	0.57–0.69	7 of 7
Natural Questions	open-domain QA	−0.54	+0.54	0.51–0.62	7 of 7
TREC-COVID	biomedical	−0.32	+0.39	(saturated)	—
legal (case study)	—	—	—	0.58	1 of 1

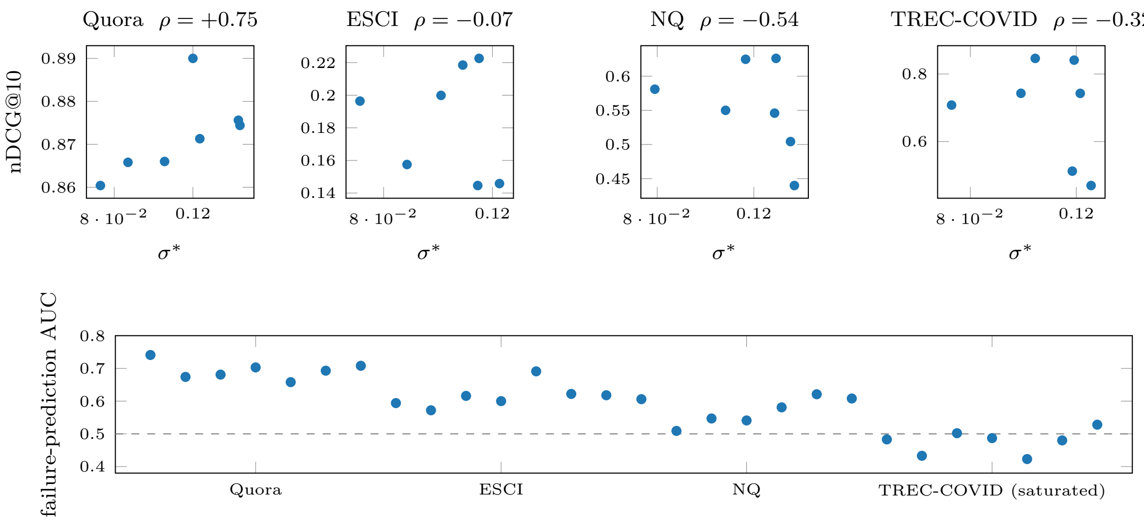

Figure 16: Every grid cell, plotted. Top: within each corpus, the seven encoders' σ^* against their nDCG@10 — rank correlation is strongly positive only where relevance is geometric matching (Quora), null on product search, and negative where semantic competence gaps dominate. Bottom: the per-document collision field's failure-prediction AUC for all 28 cells, grouped by corpus, against the chance line — above 0.5 in every informative cell (one whose failure label varies within the cell); the TREC-COVID group is uninformative because its outcome label saturates (98% of judged documents past rank 10).

One pipeline check: BGE-large's nDCG@10 on TREC-COVID reproduces its published benchmark value. Table 13 and Figures 16–17 present every cell; the grid's two findings are elaborated below.

Reading Figures 16–17. In the scatter row, the claim under test is a slope: if geometric headroom explained retrieval quality, every panel would rise to the right. Only the duplicate-matching panel does, and there the conventional summary (mean cosine) points the *wrong* way — the one setting where geometry is the task is also the one where the popular statistic misleads. The AUC strip moves the same question inside each corpus, where semantic align-

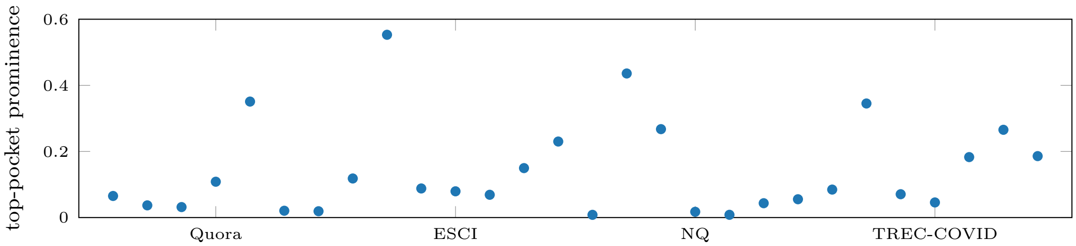

Figure 17: Structure across the grid: the most prominent pocket's merge-tree prominence for every corpus–encoder cell (all 28 exceed the null-calibrated floor). The spread within each corpus is as large as the spread across corpora — encoders disagree strongly about which corpora form tight substructure, which is precisely why pocket findings must be read per corpus–encoder pair rather than as corpus properties.

ment is held fixed by construction (one encoder, one corpus, judged documents only): there the geometric signal is positive in every informative cell. Together the two views separate the instrument's two granularities: across encoders, geometry is one factor of two and predicts only where the other is constant; within an embedding, geometry is the only free variable and predicts everywhere measured. The prominence strip adds the structural counterpart: which corpus forms pockets is a property of the corpus–encoder *pair* (the spread within a column rivals the spread across columns), which is why the instrument reports structure per embedding rather than labeling corpora as intrinsically clean or dirty.

The per-item claim is confirmed across corpora. We call a cell's outcome *informative* when the failure label actually varies — some judged documents fail at rank 10 and some do not; a label that is nearly constant cannot be predicted by anything. In every informative cell, expected-collision counts predict which judged-relevant documents fail at their own queries. The informative cells comprise all seven encoders on each of the three informative corpora — duplicate questions, product search, and open-domain QA — plus the legal case study; the AUC ranges per corpus are the final columns of Table 13. Every one of these cells is positive (sign test $p \approx 2 \times 10^{-7}$), including an AUC of 0.69 on product search for the case-study encoder despite an 84% base failure rate. The biomedical collection is uninformative for this outcome by construction, not by counterexample: with 50 queries carrying hundreds of relevant documents each, 98% of judged documents exceed rank 10 and the failure label saturates entirely.

The corpus-level claim earns a narrow scope, not a blanket confirmation. Within-corpus, across encoders, the rank correlation of σ^* with nDCG@10 is +0.75 only on duplicate matching — the one task where relevance *is* geometric proximity — and there the naive mean-cosine control is *inverted* (−0.79). On product search the correlation is null (−0.07; graded shopping relevance mixes attribute matching with intent the geometry cannot see), and on open-domain QA and biomedical search it reverses (−0.54, −0.32): the smallest encoder attains the best geometry and the worst retrieval. The mechanism is the alignment–uniformity decomposition observed directly. Retrieval quality factors into two ingredients: semantic alignment (does the encoder place related texts near each other?) and geometric headroom (does the space leave room to tell neighbors apart?). `ambit` is unsupervised by construction, so it measures only the second. σ^* therefore ranks encoders only where alignment

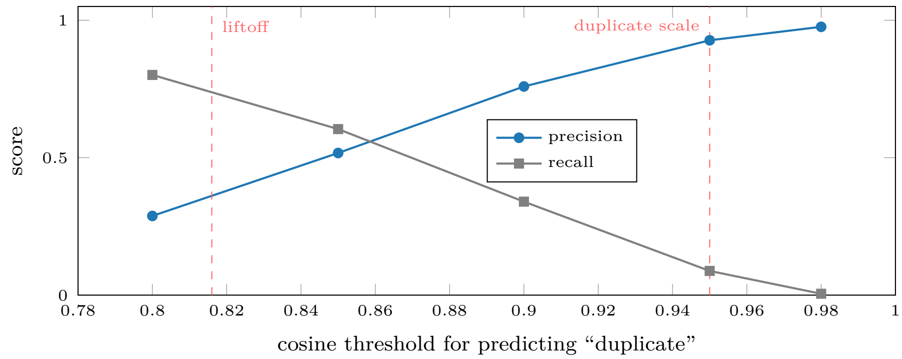

Figure 18: Duplicate prediction from pair cosine alone on Quora's annotated duplicate pairs (case-study encoder; 15,675 gold pairs). Precision rises monotonically toward the duplicate scale while recall falls; the dashed lines mark the two scales — liftoff and duplicate boundary — whose contrast the text interprets.

is essentially constant across them, and a semantically weak encoder can attain excellent geometry by spreading apart documents it does not understand. The operational conclusion: the corpus-level σ^* is a *budget* for comparative use — the same corpus under encoder variants, repair, or drift, as the guardrail gate used it — not a leaderboard; the externally validated cross-encoder readout is the per-item collision field above.

Ground truth for the scale semantics. On the duplicate-question collection, whose judgments annotate true duplicates, predicting "duplicate" from pair cosine alone yields precision that rises monotonically along the threshold ladder (0.29 at $\cos 0.80$, 0.93 at 0.95, 0.98 at 0.98 for the case-study encoder), while recall at the duplicate scale is low because annotated duplicates are mostly paraphrases living in the confusable window. This is item-level confirmation of the two-scale reading — confusable onset versus duplicate boundary — that the routing experiment (§10.5) turned into opposite repair prescriptions. Figure 18 plots the ladder.

12 Limitations and Future Work

External validation. A first cross-corpus slice now exists (§11): the per-item claim is confirmed in every informative cell and the corpus-level claim received a measured scope law. The full preregistered study remains open — more collections (one large multi-hop corpus is deferred with its queue preserved), encoder-degradation ladders, preregistration, and independent replication — and the scope law itself is a hypothesis to re-test: the alignment–uniformity split predicts that adding any alignment probe would restore ranking power on knowledge-limited tasks, but such a probe requires supervision the tool deliberately excludes. The confusion transform makes one falsifiable prediction that its near-redundant sibling does not (Table 6): duplicate pathology outranks diffuse collapse. **Query distributions.** Accepting an optional *unlabeled* query set would replace the corpus-as-queries assumption with a measured cross-set confusion, within the tool's supervision constraint. **Sharper aggregates.** The union bound is safe but loose where hubs live; Gaussian comparison inequalities may yield two-sided estimates. **Recoverability.** "Resolution headroom" — the fraction of measured crowding removable by a budgeted transform class, computed rather than predicted — would attribute measured crowding to the embedding model versus the data themselves. **Streaming structure.** The pair curve sketches trivially; the merge tree does not; a mergeable approximate hierarchy with prominence error bounds would extend the structural layer past the reservoir.

13 Conclusion

Crowding is a failure mode that embedded systems exhibit without overt errors, and it is exactly the property that global averages cannot see and binned displays cannot be trusted to measure. The instrument described here rests on one methodological commitment applied uniformly: *read continuous objects at every scale against stated nulls, and descend to named items.* The mathematics is classical — Ripley's curve, Stolarsky's identity, the distance to a measure, the condensed cluster tree, a Gaussian tail bound — and the contribution is their selection, calibration, and composition into a system with stated guarantees and stated scope, traced end to end through a single benchmark whose planted pathology every layer detects, localizes, prices, and (in the training layer) partially repairs. The negative results are reported alongside the positive ones because an instrument's reliability rests in part on a precise statement of its limits.

Appendix A Engineering Notes

The implementation is organized so that its asymptotic costs match the claims of §9: one streaming pass over the corpus, followed by statistics whose cost is independent of corpus size. This appendix records the implementation decisions behind the measured constants of Table 8.

Streaming scan. Ingestion is a single sequential pass over raw `.npy` files (memory-mapped) or sharded Parquet (chunked decode), in batches of 5×10^4 rows. Each batch of b rows updates the running mean, the norm moments, and the $d \times d$ second-moment scatter through one $X_b^\top X_b$ GEMM, and feeds a bounded reservoir sample; peak memory is $O(d^2 + \text{reservoir})$ irrespective of corpus size, and the covariance spectrum is extracted once at the end of the pass. An optional approximate mode terminates the pass after a prescribed number of rows and estimates the spectrum from that prefix; §9 documents the failure mode this introduces on sorted shards.

Blocked kernels at reservoir scale. No computation materializes an $m \times m$ matrix for reservoir size m. Top-k cosine values (the input to the distance-to-measure field) and collision counts are computed from $b \times m$ Gram tiles with partial-selection top-k (`argpartition`, linear per row, in place of a full sort). The mutual-reachability minimum spanning tree uses a vectorized Prim iteration that maintains a frontier-distance array: $O(m^2)$ arithmetic executed as full-array operations, $O(m)$ memory beyond the current tile. The structural layer additionally caps its inputs (6,000 points for the DTM field, 4,096 for the merge tree), which fixes the corpus-size-independent constants observed in Table 8.

Null sampling without vector synthesis. All isotropic-null quantities exploit the closed form for the pair cosine of independent uniform directions on S^{d-1}, $c = 2B - 1$ with $B \sim \mathrm{Beta}\big(\frac{d-1}{2}, \frac{d-1}{2}\big)$: null pair samples are drawn directly from this scalar law, so a 199-replicate rank envelope requires 199 scalar-array draws rather than 199 synthetic corpora. The anisotropy-conditioned reference uses the rotation invariance of the pair-cosine law under the angular central Gaussian — only the covariance spectrum enters — so reference pairs are generated from a small pool of spectrum-weighted Gaussian directions with no eigenvector computation. The Gaussian confusion law is evaluated with a vectorized rational approximation to erf (absolute error below 1.5×10^{-7}), which keeps the core free of dependencies beyond `numpy`.

One shared pair sample. A single random-pair cosine sample (default 2×10^5 pairs) is drawn once per corpus and reused by every pairwise statistic — $K(t)$, the envelope test, the discrepancy z, the confusion functional $C(\sigma)$, and σ^*. The continuous layer therefore costs one pass over a fixed-size array per statistic, and all reported statistics refer to the same measured sample.

Distributed training and pair-batch statistics. The encoder-loop trainer (§10.3) runs data-parallel: each rank embeds its own rows and the embeddings are gathered across ranks (the local slice retaining its gradient) before the pair losses, so pair statistics scale with world size at fixed per-device memory. This matters beyond throughput: the pair losses see only within-batch pairs, and small pair batches permit batch-level overfitting — the model separates the pairs it is shown while degrading out-of-batch structure, a failure the loop's held-out measurements catch. Gathered batches must be identically shaped on every rank; the batch builder emits a fixed count of distinct rows per rank for this reason.

Optional acceleration tiers. The core is numpy-only, with the heavy kernels routed through multi-threaded BLAS. An optional module provides the same three kernels — covariance accumulation, pair cosines, and exact blocked kNN — on torch devices (CPU, CUDA, or Apple mps, with automatic selection), transferring each chunk to the device as it streams; an approximate-neighbor path through FAISS is available where installed. These tiers change constants, not asymptotics; every number reported in this paper was produced by the numpy-only configuration (Appendix C).

Appendix B The Report Artifact

The toolkit delivers its measurements as a single self-contained, theme-adaptive HTML report with no external requests. We document its design here for completeness and reproducibility; *the artifact itself is engineering, not an academic contribution* — the contribution of this report is the measurement layer of §6–§7, which is exposed as library functions independent of any rendering. Two of the artifact's design rules are nonetheless direct consequences of the measurement theory and are worth stating. *Reference, not threshold*: every panel draws its null, because §2 established that absolute cosine is not self-interpreting. *Display versus measurement*: 2-D projections lay figures out but never carry a statistic, because §5 established that projected density is not corpus density. The remaining choices are presentation craft: a single accent color with meaning carried by position along annotated axes; figures ordered as a narrative (see it; how crowded, at what scale; who is affected; what structure it forms; why the space behaves so); and an in-place explainer on every panel stating its unstated context (null construction, subsample sizes, id provenance). One legibility finding from the artifact's development is recorded because it reinforces the display/measurement separation: a minimum-spanning-tree view drawn over the projection — even restricted to tighter-than-bulk bridges — remained too busy to read on real corpora and was demoted to a hidden option; the condensed tree of §6.4 carries the same object legibly as prominence bars.

Appendix C Test Environments

Three Linux hosts, described by their properties.

- *Measurement host* (validation battery, scaling study): a 16-core/32-thread x86-64 workstation, 123 GB RAM, NVMe storage with $\approx$ 2 GB/s measured streaming reads, Python 3.13, numpy 2.5 with multi-threaded OpenBLAS, and the *numpy-only core install* of the artifact — deliberately, since that is the configuration the portability claim is about.

- *Case-study host*: a 6-core/12-thread mobile-class x86-64 machine with 27 GB RAM, exercising the optional tabular-IO path (sharded Parquet with decode overhead).
- *Accelerated host* (encoder fine-tuning, corpus embedding for the loop, the guardrail arms, and the external grid; the exact per-document collision fields at million-row scale): a 48-core x86-64 workstation, 246 GB RAM, four workstation GPUs with 96 GB memory each. Embedding served through an OpenAI-compatible inference server or computed in-process; distributed fine-tuning over two GPUs with gradients synchronized per step.

Every reported *measurement* is computed by the released `numpy` core regardless of host; the accelerated host contributes embedding, training, and the large exact collision fields (an optional `torch` path in the artifact). No experiment uses code outside the artifact and its stated experiment scripts; random seeds are fixed and stated per experiment.

Appendix D Formula Glossary

Every formula in the report is restated here in plain language: what it formulates, why it matters, how it is used, and what every symbol and operation means. Entries follow the order of first appearance. The first entry explains the shared notation that recurs throughout; each later entry still lists all of its own symbols.

How to read the notation (used throughout).

- n — the number of documents in the corpus. d — the number of coordinates in each embedding vector (the "dimension").
- i, j — indices naming individual documents, each running from 1 to n. Writing x_i means "the embedding vector of document number i"; a pair (i, j) names two different documents.
- $\Pr[\,\text{event}\,]$ — "the probability that the statement inside the brackets is true." A subscript names what is random: in $\Pr_{(i,j)}[\cdots]$ the random thing is the choice of pair — draw two distinct documents at random, and ask how often the bracketed statement holds. With no subscript, the randomness is stated in the entry (for the flip law it is the query noise).
- $\{\sigma : C(\sigma) \leq 1\}$ — "the set of all values σ such that the condition after the colon holds"; $\max\{\cdots\}$ takes the largest member of a set.
- $\sum_{j \neq i}$ — add up the expression that follows, once for every allowed value of the index (j running over all documents except i). $\frac{1}{n^2}\sum_{i,j}$ — add over all n^2 ordered pairs, then divide by their count: an average over pairs.
- $\int f(r)\, dK(r)$ — an average of f weighted by how often each distance r occurs among pairs; "$dK(r)$" means "the fraction of pairs whose distance is (approximately) r."
- $\mathbb{E}$ — the expected value (average) over the stated randomness.
- $\log$ — the natural logarithm; e^x — the exponential function, its inverse.

- $\partial f/\partial r$ — the derivative of f with respect to r: how fast f changes when r changes a little. $\propto$ — "proportional to": equal up to a constant factor that does not depend on the variable under discussion.
- $\|v\|$ — the Euclidean length of a vector v (square root of the sum of its squared coordinates). $\langle u, v\rangle$ — the dot product of two vectors (multiply matching coordinates, add them up).
- Φ — the standard normal cumulative distribution function: $\Phi(u)$ is the probability that a standard Gaussian (bell-curve) random number falls below u. $\Phi(0) = 1/2$; $\Phi(-4) \approx 3 \times 10^{-5}$.

Unit vectors and cosine similarility: $c_{ij} = \langle x_i, x_j\rangle$, points on $\mathbb{S}^{d-1}$.

- **What it says:** Every document is represented as a length-one arrow in d-dimensional space, and the similarity of two documents is the cosine of the angle between their arrows.
- **Why it matters:** Cosine is the score retrieval systems actually rank by, so all geometry in the report is expressed in the same units the system uses.
- **How it is used:** Every statistic in the report is a function of these pairwise cosines.
- **Term by term:**
  - x_i — the embedding vector of document i, rescaled to length 1 (each of its d coordinates is a number; the whole vector is one point in d-dimensional space).
  - $\mathbb{S}^{d-1}$ — the unit sphere in d dimensions: the set of all vectors of length exactly 1. The superscript $d-1$ is the standard name for the sphere's surface (a globe in 3-D space has a 2-D surface); nothing in the report depends on this convention.
  - $\langle x_i, x_j\rangle$ — the dot product of the two vectors, which for length-one vectors equals the cosine of the angle between them.
  - c_{ij} — shorthand for that cosine between documents i and j: $+1$ means identical direction, 0 unrelated (perpendicular), -1 opposite.

Chord distance: $r_{ij} = \|x_i - x_j\| = \sqrt{2 - 2c_{ij}}$.

- **What it says:** The straight-line distance between two points on the sphere, computed directly from their cosine.
- **Why it matters:** The query-noise law below is exact in chord distance, not in cosine, so the report converts between the two constantly; this identity is the dictionary.
- **Term by term:**
  - $x_i - x_j$ — the vector pointing from document j's position to document i's.
  - $\|x_i - x_j\|$ — that vector's length: the ordinary straight-line distance between the two points.

- r_{ij} — shorthand for that distance.
- $\sqrt{2 - 2c_{ij}}$ — the same distance computed from the cosine alone. It follows by expanding the squared length: $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle = 1 + 1 - 2c_{ij}$.
- Reference points: identical items have $r = 0$; perpendicular ones $r = \sqrt{2} \approx 1.41$; opposite ones $r = 2$.

The null cosine scale: coordinates $\sim \mathcal{N}(0, 1/d)$, typical cosine $\approx 1/\sqrt{d}$.

- **What it says:** If points are spread uniformly over a high-dimensional sphere, each coordinate behaves like a Gaussian random number with variance $1/d$, and the cosine between two random points is almost always within a few multiples of $1/\sqrt{d}$ of zero (at $d = 768$, about ± 0.036).
- **Why it matters:** This is the baseline against which "suspiciously similar" is judged: in high dimension, unrelated items are *very* nearly perpendicular, so even modest cosines are extremely surprising under the null.
- **Term by term:**
  - $\sim$ — "is distributed as": the quantity on the left behaves like a random draw from the distribution on the right.
  - $\mathcal{N}(0, 1/d)$ — the normal (bell-curve) distribution with mean 0 and variance $1/d$: values concentrate near zero, spread $\approx 1/\sqrt{d}$.
  - $1/\sqrt{d}$ — the typical size of a null cosine; it shrinks as dimension grows, which is the concentration phenomenon.

Pair-closeness curve: $K(t) = \Pr_{(i,j)}[\, c_{ij} \geq t \,]$.

- **What it says:** For every similarity threshold t simultaneously, the fraction of document pairs that are at least t-similar.
- **Why it matters:** It is the report's primary object: a complete, bin-free record of how much of the corpus sits at every closeness scale. Crowding appears as an excess of very close pairs — the curve "lifting off" above its reference at high t.
- **How it is used:** Estimated exactly by sorting a large random sample of pair cosines; compared against the two null curves below; the comparison is tested with the rank-envelope test.
- **Term by term:**
  - K — the name of the curve (after Ripley's K function in spatial statistics, of which this is the spherical, cosine-coordinate version). For each input t it returns one number between 0 and 1.
  - t — a similarity threshold, swept continuously from low to high; the curve is read at every t at once, which is what "no bins" means.

- (i, j) — a pair of two distinct documents, chosen uniformly at random from the corpus.
- $\Pr_{(i,j)}[\, c_{ij} \geq t\,]$ — the probability, over that random choice of pair, that the pair's cosine reaches the threshold — equivalently, simply the fraction of all pairs that do.

The uniform null for pair cosines: $c = 2B - 1$ **with** $B \sim \mathrm{Beta}\big(\frac{d-1}{2}, \frac{d-1}{2}\big)$**.**

- **What it says:** If points were spread perfectly evenly on the sphere, the cosine of a random pair would follow this exact, known distribution — a Beta distribution stretched from $[0, 1]$ to $[-1, 1]$.
- **Why it matters:** It gives the "perfectly spread corpus" reference curve in closed form: no simulation of fake corpora is needed (one draws random *numbers*, not random *vectors*), which is what makes 199-replicate null bands cheap.
- **Term by term:**
  - c — the pair cosine whose null distribution is being stated.
  - B — a random draw from the Beta distribution, which produces numbers between 0 and 1.
  - $\mathrm{Beta}(a, a)$ with $a = (d-1)/2$ — a two-parameter family of distributions on $[0, 1]$; with both parameters equal it is a symmetric bell around 1/2 that narrows as dimension grows (the concentration phenomenon again).
  - $2B - 1$ — rescales $[0, 1]$ onto $[-1, 1]$ so the result is a cosine: $B = 1/2$ (the typical draw) maps to $c = 0$, i.e. perpendicular.

The anisotropy-matched reference (ACG): $x = \sqrt{\lambda} \odot g \,/\, \|\sqrt{\lambda} \odot g\|$**.**

- **What it says:** A second reference corpus that copies the real corpus's overall *shape* — its elongation along some directions and flattening along others — but contains no clustering whatsoever.
- **Why it matters:** Learned embeddings almost always occupy a narrow cone (the "cone effect"), which inflates all cosines benignly. Comparing the data against this reference, instead of against the uniform null alone, separates crowding the cone explains (benign) from crowding it cannot (genuine pockets).
- **Term by term:**
  - g — a vector of d independent standard Gaussian random numbers: a direction chosen with no preference at all.
  - λ — the corpus's covariance eigenvalues: a list of d non-negative numbers, one per principal direction, each saying how much variance the corpus has along that direction. Together they describe the corpus's ellipsoidal shape.

  - $\sqrt{\lambda} \odot g$ — multiply each coordinate of g by the square root of the matching eigenvalue ($\odot$ is coordinate-wise multiplication); this stretches the isotropic random cloud into the corpus's ellipsoid.
  - dividing by $\|\cdots\|$ — rescale to length one, so the reference point lands on the sphere like the real data.
  - Only the eigenvalues enter — rotating a corpus changes none of its pairwise cosines, so the eigen*vectors* are irrelevant here.

The global rank-envelope test and the liftoff cosine.

- **What it says:** To decide whether the data curve $K(t)$ genuinely exceeds the null band — across *all* thresholds at once, not cherry-picking one — the test simulates many null curves, ranks the data curve among them at every threshold simultaneously, and returns one global p-value. The *liftoff cosine* is the smallest similarity at which the data curve exits the envelope, reported only when the global test rejects.

- **Why it matters:** Reading exceedance off a pointwise band at each threshold separately commits a multiple-comparison error — on pure noise it fires essentially always (measured: every one of 40 null corpora). The global test's false alarms match its nominal rate.

- **How it is used:** Every liftoff claim in the report is gated on this global p; extreme-rank-length (ERL) ordering breaks ties among simulated curves.

- **Term by term:**
  - envelope — the band spanned by the simulated null curves (here 199 of them, drawn via the Beta law above).
  - p-value — the probability, if the corpus were truly well-spread, of seeing a curve as extreme as the data's anywhere along its length. Small p (e.g. 0.005) means "a well-spread corpus essentially never looks like this." With 199 simulations the smallest attainable p is $1/200 = 0.005$.
  - ERL (extreme rank length) — the tie-breaking rule: when two curves are equally extreme at their worst point, the one that stays extreme over a longer stretch ranks as more extreme.
  - liftoff cosine — the onset similarity of genuine tight-pair structure: below it the data track the reference; above it they exceed the envelope.

Stolarsky invariance and the occupancy discrepancy z: cap discrepancy $=$ const $-\frac{1}{n^2}\sum_{i,j}\|x_i - x_j\|$.

- **What it says:** Take every possible "cap" on the sphere (every center, every radius), compare how many points each cap holds against how many it should hold, square the errors, and add them all up. Stolarsky's theorem says this total — which looks uncomputable — equals a constant minus the average distance between pairs of points.

- **Why it matters:** It is the exact completion of what binned occupancy displays approximate: *every* lattice at *every* scale simultaneously, with no lattice to choose. A corpus that under-spreads (crowds anywhere, at any scale) has a smaller average pairwise distance.

- **How it is used:** The report computes the mean pairwise chord and standardizes it against the uniform null, giving the occupancy discrepancy z. On the benchmark, uniform controls score $z \approx 0$ and the planted pocket scores $z = -47$.

- **Term by term:**
  - cap discrepancy — the total squared error between actual and ideal cap occupancy, accumulated over all cap centers and radii; the formal object on the left of the identity.
  - const — an explicit constant that depends only on n and d, not on where the points are; it plays no role in comparisons.
  - $\frac{1}{n^2}\sum_{i,j}\|x_i - x_j\|$ — the average chord distance over all ordered pairs of documents (sum over every pair, divide by the number of pairs).
  - z — the reported score: the observed mean chord minus the null mean, divided by the null standard deviation, where the null mean and standard deviation come from redrawing the same number of pair cosines from the Beta law. $z = -47$ reads "the corpus's average distance sits 47 standard deviations of sampling noise below the well-spread value."

Distance to a measure: $\mathrm{DTM}_m(x_i) = \big(\frac{1}{k}\sum_{j=1}^{k} r_{i,(j)}^2\big)^{1/2}$ **with** $k = \lceil mn \rceil$**.**

- **What it says:** For each item: the typical distance it must reach to gather the nearest m-fraction of the corpus around it — a per-item crowding radius. Small radius = crowded; large radius = isolated.

- **Why it matters:** It descends from corpus-level statistics to *named items*, which is what makes findings actionable, and it is provably stable: outliers and small perturbations of the corpus move it only slightly (the Wasserstein-stability guarantee).

- **How it is used:** Computed for every reservoir item; its low tail (the most crowded items) is compared against the uniform corpus's band; the report lists the most-crowded and most-isolated items by identifier.

- **Term by term:**
  - $\mathrm{DTM}_m(x_i)$ — the function's name ("distance to the measure") applied to document i at mass fraction m; one number per document.
  - m — the mass fraction, the method's one knob (e.g. 0.02: gather 2% of the corpus); being a fraction makes it scale-free in corpus size.
  - $k = \lceil mn \rceil$ — that fraction converted to a count of neighbors; $\lceil\cdot\rceil$ rounds up to a whole number.

- $r_{i,(j)}$ — the distance from item i to its jth nearest neighbor; the parenthesized subscript (j) means the distances have been sorted ascending, so (1) is the closest.
- $\frac{1}{k}\sum_{j=1}^{k} r_{i,(j)}^2$ — the average of the squared distances to the k nearest neighbors.
- $(\cdots)^{1/2}$ — the square root of that average: a root-mean-square radius, an average robust to any single neighbor.

The merge tree: mutual reachability, birth, death, prominence.

- **What it says:** Connect items by their shortest bridges (single linkage) under the *mutual-reachability* distance. As the connection scale grows, tight groups form and later dissolve into the bulk. A pocket's *birth* is the scale at which it first holds together, its *death* the scale at which it merges into something larger, and its *prominence* = death − birth is how long it exists as its own entity.

- **Why it matters:** Prominence replaces every flat clustering threshold: a pocket is reported not because it passed a cutoff but because it persists across a wide range of scales. A null-calibrated prominence floor (the 99th percentile of prominence produced by pure noise at matched size and dimension) suppresses spurious pockets.

- **Term by term:**
  - single linkage — build connections by repeatedly merging the two closest groups; the merge tree records the order and scale of every merge.
  - mutual reachability of a, b — $\max\{d(a,b), \mathrm{core}(a), \mathrm{core}(b)\}$: the largest of the three listed quantities, where $d(a,b)$ is the ordinary distance between the two points and $\mathrm{core}(x)$ is the distance from x to its own kth nearest neighbor. Taking the max means two points count as close only if each is also locally dense — a noise point far from everything cannot form a spurious bridge.
  - birth / death — the connection scales at which a group first assembles and at which it is absorbed into an older group (the *elder rule*: when two groups meet, the younger one dies).
  - prominence — death minus birth, in distance units: the width of the scale range over which the pocket is a separate entity.
  - excess of mass — the selection rule choosing, on each branch of the tree, the cluster that persists longest.

The query-noise model: $q = x + \sigma g$, $g \sim \mathcal{N}(0, I_d)$.

- **What it says:** A query aimed at item x lands nearby rather than exactly on it: x plus Gaussian noise of scale σ in every direction.

- **Why it matters:** It converts geometry into operational risk. Real queries paraphrase, truncate, and typo; σ is the abstract knob for all such imprecision, and every downstream quantity is a function of it.

- **Term by term:**

– q — the query vector the system actually searches with.
– x — the item the query "means": its intended target.
– $g \sim \mathcal{N}(0, I_d)$ — a fresh standard Gaussian vector: each of the d coordinates is an independent bell-curve draw with mean 0 and variance 1 (I_d, the identity matrix, encodes "independent, unit variance, no preferred direction").
– σ — the noise scale: how far, typically, the query lands from its target, in the same units as chord distance. $\sigma = 0$ means perfect queries; larger σ means sloppier ones.

The exact flip law: $\Pr[\langle q, y\rangle > \langle q, x\rangle] = \Phi(-r/2\sigma)$**.**

- **What it says:** The probability that a noisy query aimed at x scores a competitor y higher than x itself depends on nothing but the distance between x and y, through one evaluation of the normal tail. Exactly — in any dimension.
- **Why it matters:** It is the bridge between geometry and retrieval errors, and it explains why near-duplicates are catastrophic: as $r \to 0$ the probability approaches $\Phi(0) = 1/2$ — a duplicated document loses half its queries to its twin no matter how small the noise is.
- **How it is used:** Verified by direct simulation to Monte-Carlo precision (max deviation 9×10^{-4}); summed over competitors to give $C(\sigma)$; used as the training loss kernel.
- **Term by term:**
  – x, y — the query's intended target and one competing document, both unit vectors; q — the noisy query from the previous entry.
  – $\langle q, y\rangle$ — the dot product of query and competitor: exactly the similarity score retrieval ranks by. The bracketed statement $\langle q, y\rangle > \langle q, x\rangle$ reads "the competitor scores higher than the intended target," i.e. the retrieval errs.
  – $\Pr[\cdots]$ — the probability of that error over the randomness of the query noise g.
  – r — the chord distance between x and y; σ — the query-noise scale.
  – $\Phi(-r/2\sigma)$ — the normal tail evaluated at $-r/2\sigma$. Why this argument: the set of points scoring x and y equally is the perpendicular bisector of the segment joining them, which sits at distance $r/2$ from x; Gaussian noise of scale σ crosses a boundary $r/2$ away with probability $\Phi(-(r/2)/\sigma)$. Larger separation or smaller noise pushes the argument more negative and the error probability toward zero.

Expected collisions: $C(\sigma) = \sum_{j \neq i} \Phi(-r_{ij}/2\sigma) \approx (n-1) \int \Phi(-r/2\sigma)\, dK(r)$**.**

- **What it says:** For an average item, the expected number of wrong competitors that a noisy query would rank above it: add the flip probability over every competitor.
- **Why it matters:** By Boole's inequality (the probability that *any* of several events occurs is at most the sum of their individual probabilities), $C(\sigma)$ bounds the probability of *any* retrieval error, no matter how the competitors are correlated — so it is conservative: crowding can make it loose, never wrong.

- **Term by term:**
  - $C(\sigma)$ — the function's name: expected collision count at noise scale σ; one number per σ, increasing in σ.
  - i — the item being queried; j — each competing document in turn.
  - $\sum_{j \neq i} \Phi(-r_{ij}/2\sigma)$ — the flip probability against competitor j (previous entry), added up over all $n-1$ competitors.
  - $(n-1)\int \Phi(-r/2\sigma)\, dK(r)$ — the same sum computed from the pair-distance distribution alone: average the flip probability over the distribution of pair distances (that is the integral against dK), then multiply by the number of competitors. This is why a random pair *sample* suffices — no per-item computation is needed for the corpus-level curve.

Resolution bandwidth: $\sigma^* = \max\{\sigma : C(\sigma) \leq 1\}$.

- **What it says:** The largest amount of query noise the corpus can absorb before, in expectation, one wrong item outranks the right one.
- **Why it matters:** It is the report's headline operational scalar: a noise *budget* in retrieval units. Comparing σ^* against the same corpus size spread uniformly (the ratio reported alongside it) says what fraction of the theoretically available budget the embedding retains.
- **Term by term:**
  - $\{\sigma : C(\sigma) \leq 1\}$ — the set of all noise scales at which the expected collision count stays at or below one; max takes the largest such scale.
  - the threshold 1 — "one expected collision per item": the point where retrieval starts, on average, to lose. It is a convention (a tolerance), and the report states it wherever σ^* appears.
  - C is monotone — more noise always means more collisions — so this largest admissible σ is well defined and is found by bisection: repeatedly halve an interval known to contain the crossing.

Uniformity as a collision bound: $U_t = \log \mathbb{E}\, e^{-t\|x-y\|^2}$ and $C(\sigma) \leq (n-1)\, e^{U_t}$ at $t = 1/(8\sigma^2)$.

- **What it says:** The widely used Wang–Isola "uniformity" loss — the log of the average of e^{-tr^2} over pairs — is, at the right temperature, exactly the logarithm of a Chernoff upper bound on the collision count $C(\sigma)$.
- **Why it matters:** It gives the popular loss operational units: its temperature knob t is not folklore but a choice of query-noise scale ($t = 2$ corresponds to $\sigma = 0.25$). It also explains when uniformity misleads — it is a bound, and the report measures where the bound's ordering and the exact functional's ordering disagree.
- **Term by term:**

– U_t — the uniformity functional's value at temperature t (one number per t; more negative = more spread).
– x, y — a random pair of documents; $\mathbb{E}$ — the average over that random pair.
– $e^{-t\|x-y\|^2}$ — a Gaussian-shaped penalty: 1 when the pair coincides, decaying with squared distance at rate t (large t = only very close pairs are penalized).
– log — the natural logarithm of the average, following the original definition.
– $t = 1/(8\sigma^2)$ — the dictionary between temperature and noise scale; where the identity holds.
– the proof, in one line — the normal tail bound $\Phi(-z) \le e^{-z^2/2}$ (a Chernoff bound) applied to $z = r/2\sigma$ turns each flip probability into $e^{-r^2/8\sigma^2}$, and summing gives $(n-1)\, e^{U_t}$.

The global suite: mean pair cosine, IsoScore, effective rank, hubness skew.

- **What it says:** Four whole-corpus summaries used as context and as baselines. Mean pair cosine: the average c_{ij} over random pairs (how coned the corpus is). IsoScore: how close the covariance is to perfectly isotropic — equal variance in all directions — on a 0-to-1 scale. Effective rank: exp of the entropy of the normalized eigenvalue distribution — roughly, how many directions carry meaningful variance (between 1 and d). Hubness skew: the skewness (asymmetry) of the count of how often each item appears in others' k-nearest-neighbor lists — heavy right skew means a few "hub" items intrude into everyone's neighborhoods.
- **Why it matters:** These are the statistics that practitioners already consult; the detection-power study shows them blind to small pockets that the continuous layer detects at full power, because they are global averages.

Training losses: $\mathcal{L}_{\text{conf}} = \text{mean}_{(i,j)\notin\mathcal{E}}\, \Phi\big(-\|z_i - z_j\|/2\sigma\big)$ and $\mathcal{L}_{\text{pres}} = \text{mean}_i\, \text{KL}\big(p_i^{\text{base}} \,\|\, p_i\big)$.

- **What it says:** Two differentiable objectives built from the measurements. The first is the batch version of the collision count: the average flip probability over batch pairs, at the *measured* noise scale, excluding a set of protected pairs. The second pins local semantics: for each anchor, the distribution "who in this batch is similar to me" under the model being trained must stay close to the same distribution under the frozen base model.
- **Why it matters:** $\mathcal{L}_{\text{conf}}$ spends gradient only where confusion actually lives, and $\mathcal{L}_{\text{pres}}$ prevents the repair from destroying meaning. Their balance (λ_p) is the loop's central tension: recovery against fidelity.
- **Term by term:**
  – $\mathcal{L}$ — a loss: a number training tries to make small; the subscripts name the two losses (confusion, preservation).
  – z_i — the embedding of batch item i under the model currently being trained (it changes every step; x_i is reserved for the fixed corpus vectors).

- $\text{mean}_{(i,j)\notin\mathcal{E}}$ — average over all ordered batch pairs *except* those in the excluded set; $\notin$ means "not a member of."
- $\mathcal{E}$ — the excluded pairs: known positives and the false-negative guard (each anchor's top-m base-model neighbors, which are likely unlabeled true relatives and must not be pushed apart).
- $\Phi(-\|z_i - z_j\|/2\sigma)$ — the flip law applied to the batch pair at the measured scale $\sigma = \sigma^*$.
- $\mathrm{KL}(p \,\|\, q)$ — the Kullback–Leibler divergence: the standard measure of how much a distribution q misrepresents a reference distribution p; zero exactly when they agree, larger the more they differ. The double bar $\|$ is part of the notation and separates reference from candidate.
- p_i and p_i^{base} — for anchor i, the softmax over its cosines to the rest of the batch (exponentiate each cosine divided by a temperature of 0.05, then normalize to sum to one — low temperature sharpens the distribution toward the nearest neighbors), computed under the trained model and under the frozen base model respectively.

Gradient locality: $\partial\Phi(-r/2\sigma)/\partial r \propto e^{-r^2/8\sigma^2}$.

- **What it says:** The derivative of the flip probability with respect to pair distance dies off as a Gaussian in r: pairs farther apart than about 3σ receive exponentially negligible gradient.
- **Why it matters:** It is a structural safety guarantee: minimizing $\mathcal{L}_{\text{conf}}$ *cannot* disturb the corpus's far-pair (dissimilarity) structure, by construction rather than by tuning. The test suite enforces it (a near pair receives more than 10^3 times the gradient of an orthogonal pair).
- **Term by term:**
  - $\partial\Phi(-r/2\sigma)/\partial r$ — how fast the flip probability changes as the pair distance r changes: exactly the signal backpropagation sends to a pair during training.
  - $\propto$ — proportional to: equal up to a constant that does not involve r.
  - $e^{-r^2/8\sigma^2}$ — a Gaussian in r: near 1 for close pairs, and collapsing extremely fast once r exceeds a few multiples of σ — at $r = 3\sigma$ it is already below $e^{-9/8} \approx 0.32$ of its peak and falling as the *square* of the distance in the exponent.

Neighbor overlap at k.

- **What it says:** For each item, the fraction of its k nearest neighbors that two embeddings (base and tuned) have in common; the report uses $k = 10$ and averages over held-out items.
- **Why it matters:** It is the retrieval-relevant measure of "did the model change who is similar to whom" — comparing neighbor *identities*, which global similarity statistics can miss — and it is the loop's licensing criterion (≥ 0.90).

- **Term by term:**
  - k — the neighborhood size: how many nearest neighbors each item's list contains (10 here, matching a retrieval page).
  - overlap for one item — $|A \cap B|/k$, where A and B are the item's neighbor sets under the two embeddings, $\cap$ is the set intersection (members common to both), and $|\cdot|$ counts members: 1 means identical neighborhoods, 0 fully reshuffled.

Significance machinery of the loop section.

- **McNemar's exact test** asks whether a model change helped or hurt on *paired* outcomes: it looks only at the discordant queries (helped by one model, failed by the other — here 118 against 131) and asks whether that split is farther from 50:50 than coin flips would produce.

- **The paired sign test** does the same for per-item quantities: of 408 held-out flagged items, 406 improved — the probability of a split that lopsided under "no effect" is astronomically small ($p \approx 10^{-118}$), which is how a tiny median effect can still be unambiguous.

- **Bootstrap confidence intervals** quantify uncertainty without distributional assumptions: resample the data with replacement many times, recompute the statistic each time, and read the spread; an interval containing zero means the sign of the effect is not established.

- **AUC** (area under the ROC curve, equivalently the Mann–Whitney statistic) measures a predictor's ability to rank: the probability that a randomly chosen failing document scores higher on the predictor than a randomly chosen non-failing one — 0.5 is chance, 1.0 is perfect ordering.

References

[1] J. Gao, D. He, X. Tan, T. Qin, L. Wang, T.-Y. Liu. Representation degeneration problem in training natural language generation models. *ICLR*, 2019.

[2] K. Ethayarajh. How contextual are contextualized word representations? *EMNLP*, 2019.

[3] J. Mu, S. Bhat, P. Viswanath. All-but-the-top: simple and effective postprocessing for word representations. *ICLR*, 2018.

[4] X. Cai, J. Huang, Y. Bian, K. Church. Isotropy in the contextual embedding space: clusters and manifolds. *ICLR*, 2021.

[5] O. Roy, M. Vetterli. The effective rank: a measure of effective dimensionality. *EUSIPCO*, 2007.

[6] W. Rudman, N. Gillman, T. Rush, C. Eickhoff. IsoScore: measuring the uniformity of embedding space utilization. *Findings of ACL*, 2022.

[7] T. Wang, P. Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. *ICML*, 2020.

[8] M. Radovanović, A. Nanopoulos, M. Ivanović. Hubs in space: popular nearest neighbors in high-dimensional data. *JMLR* 11, 2010.

[9] Q. Garrido, R. Balestriero, L. Najman, Y. LeCun. RankMe: assessing the downstream performance of pretrained self-supervised representations by their rank. *ICML*, 2023.

[10] K. Agrawal, A. Mondal, A. Ghosh, B. Richards. α-ReQ: assessing representation quality by measuring eigenspectrum decay. *NeurIPS*, 2022.

[11] A. Tsitsulin, M. Munkhoeva, B. Perozzi. Unsupervised embedding quality evaluation. *TAG-ML @ ICML*, 2023.

[12] D. Friedman, A. B. Dieng. The Vendi score: a diversity evaluation metric for machine learning. *TMLR*, 2023.

[13] A. Pasarkar, A. B. Dieng. Cousins of the Vendi score: a family of similarity-based diversity metrics. *AISTATS*, 2024.

[14] L. G. S. Giraldo, M. Rao, J. C. Príncipe. Measures of entropy from data using infinitely divisible kernels. *IEEE Trans. Information Theory* 61(1), 2015.

[15] A. Ospanov, F. Farnia. Do Vendi scores converge with finite samples? Truncated Vendi score for finite-sample convergence guarantees. arXiv:2410.21719, 2024.

[16] B. D. Ripley. The second-order analysis of stationary point processes. *J. Applied Probability* 13(2), 1976.

[17] J. Besag. Contribution to the discussion of Dr Ripley's paper. *JRSS-B* 39(2), 1977.

[18] S. M. Robeson, A. Li, C. Huang. Point-pattern analysis on the sphere. *Spatial Statistics* 10, 2014.

[19] A. J. Baddeley, J. Møller, R. Waagepetersen. Non- and semi-parametric estimation of interaction in inhomogeneous point patterns. *Statistica Neerlandica* 54(3), 2000.

[20] D. E. Tyler. Statistical analysis for the angular central Gaussian distribution on the sphere. *Biometrika* 74(3), 1987.

[21] M. Myllymäki, T. Mrkvička, P. Grabarnik, H. Seijo, U. Hahn. Global envelope tests for spatial processes. *JRSS-B* 79(2), 2017.

[22] K. B. Stolarsky. Sums of distances between points on a sphere II. *Proc. AMS* 41(2), 1973.

[23] J. S. Brauchart, J. Dick. A simple proof of Stolarsky's invariance principle. *Proc. AMS* 141, 2013 (arXiv:1101.4448).

[24] C. E. Gehlke, K. Biehl. Certain effects of grouping upon the size of the correlation coefficient in census tract material. *JASA* 29(185A), 1934.

[25] S. Openshaw, P. J. Taylor. A million or so correlation coefficients: three experiments on the modifiable areal unit problem. In *Statistical Applications in the Spatial Sciences*, Pion, 1979.

[26] S. Openshaw. *The Modifiable Areal Unit Problem.* CATMOG 38, Geo Books, 1984.

[27] W. S. Robinson. Ecological correlations and the behavior of individuals. *American Sociological Review* 15(3), 1950.

[28] D. W. Scott. On optimal and data-based histograms. *Biometrika* 66(3), 1979.

[29] D. Freedman, P. Diaconis. On the histogram as a density estimator: L_2 theory. *Z. Wahrsch. Verw. Gebiete* 57, 1981.

[30] D. W. Scott. Averaged shifted histograms: effective nonparametric density estimators in several dimensions. *Annals of Statistics* 13(3), 1985.

[31] B. W. Silverman. *Density Estimation for Statistics and Data Analysis*. Chapman & Hall, 1986.

[32] F. Chazal, D. Cohen-Steiner, Q. Mérigot. Geometric inference for probability measures. *Found. Comput. Math.* 11(6), 2011.

[33] F. Chazal, B. Michel. An introduction to topological data analysis: fundamental and practical aspects for data scientists. *Frontiers in AI* 4, 2021.

[34] F. Chazal, L. J. Guibas, S. Y. Oudot, P. Skraba. Persistence-based clustering in Riemannian manifolds. *J. ACM* 60(6), 2013.

[35] R. J. G. B. Campello, D. Moulavi, A. Zimek, J. Sander. Hierarchical density estimates for data clustering, visualization, and outlier detection. *ACM TKDD* 10(1), 2015.

[36] A. Rolle, L. Scoccola. Stable and consistent density-based clustering via multiparameter persistence. *JMLR*, 2024 (arXiv:2005.09048).

[37] J. A. Hartigan. Consistency of single linkage for high-density clusters. *JASA* 76(374), 1981.

[38] K. Chaudhuri, S. Dasgupta. Rates of convergence for the cluster tree. *NeurIPS*, 2010.

[39] D. Cohen-Steiner, H. Edelsbrunner, J. Harer. Stability of persistence diagrams. *Discrete Comput. Geom.* 37, 2007.

[40] K. Beyer, J. Goldstein, R. Ramakrishnan, U. Shaft. When is "nearest neighbor" meaningful? *ICDT*, 1999.

[41] E. Levina, P. J. Bickel. Maximum likelihood estimation of intrinsic dimension. *NeurIPS*, 2005.

[42] E. Facco, M. d'Errico, A. Rodriguez, A. Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports* 7, 2017.

[43] M. Wattenberg, F. Viégas, I. Johnson. How to use t-SNE effectively. *Distill*, 2016.

[44] A. Narayan, B. Berger, H. Cho. Assessing single-cell transcriptomic variability through density-preserving data visualization. *Nature Biotechnology* 39, 2021.

[45] S. Kornblith, M. Norouzi, H. Lee, G. Hinton. Similarity of neural network representations revisited. *ICML*, 2019.

[46] V. Papyan, X. Y. Han, D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *PNAS* 117(40), 2020.

[47] Qwen Team. Qwen3 Embedding: advancing text embedding and reranking through foundation models. *arXiv:2506.05176*, 2025.

[48] S. Iyer, N. Dandekar, K. Csernai. First Quora dataset release: question pairs. Quora Engineering Blog, 2017.

[49] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, M. Zhou. MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained transformers. *NeurIPS*, 2020.

[50] K. Song, X. Tan, T. Qin, J. Lu, T.-Y. Liu. MPNet: masked and permuted pre-training for language understanding. *NeurIPS*, 2020.

[51] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, I. Gurevych. BEIR: a heterogeneous benchmark for zero-shot evaluation of information retrieval models. *NeurIPS Datasets and Benchmarks*, 2021.

[52] E. Voorhees, T. Alam, S. Bedrick, D. Demner-Fushman, W. R. Hersh, K. Lo, K. Roberts, I. Soboroff, L. L. Wang. TREC-COVID: constructing a pandemic information retrieval test collection. *SIGIR Forum* 54(1), 2020.

[53] T. Kwiatkowski et al. Natural Questions: a benchmark for question answering research. *TACL* 7, 2019.

[54] C. K. Reddy, L. Màrquez, F. Valero, N. Rao, H. Zaragoza, S. Bandyopadhyay, A. Biswas, A. Xing, K. Subbian. Shopping queries dataset: a large-scale ESCI benchmark for improving product search. *arXiv:2206.06588*, 2022.

[55] N. Reimers, I. Gurevych. Sentence-BERT: sentence embeddings using Siamese BERT-networks. *EMNLP*, 2019.

[56] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, F. Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv:2212.03533*, 2022.

[57] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff. C-Pack: packaged resources to advance general Chinese and English text embedding. *SIGIR*, 2024.

[58] L. Merrick, D. Xu, G. Nuti, D. Campos. Arctic-Embed: scalable, efficient, and accurate text embedding models. *arXiv:2405.05374*, 2024.

[59] Gemma Team. EmbeddingGemma: efficient open text embeddings (model release and technical documentation), 2025.

[60] NVIDIA. Nemotron-Pretraining-Legal-v1 (dataset, CC-BY-4.0), 2026. `https://huggingface.co/datasets/nvidia/Nemotron-Pretraining-Legal-v1`; the dataset card directs citation to the NVIDIA Nemotron 3 Ultra technical report.